

BFS Traversal

Step 1: Start at E

Legend

Complete

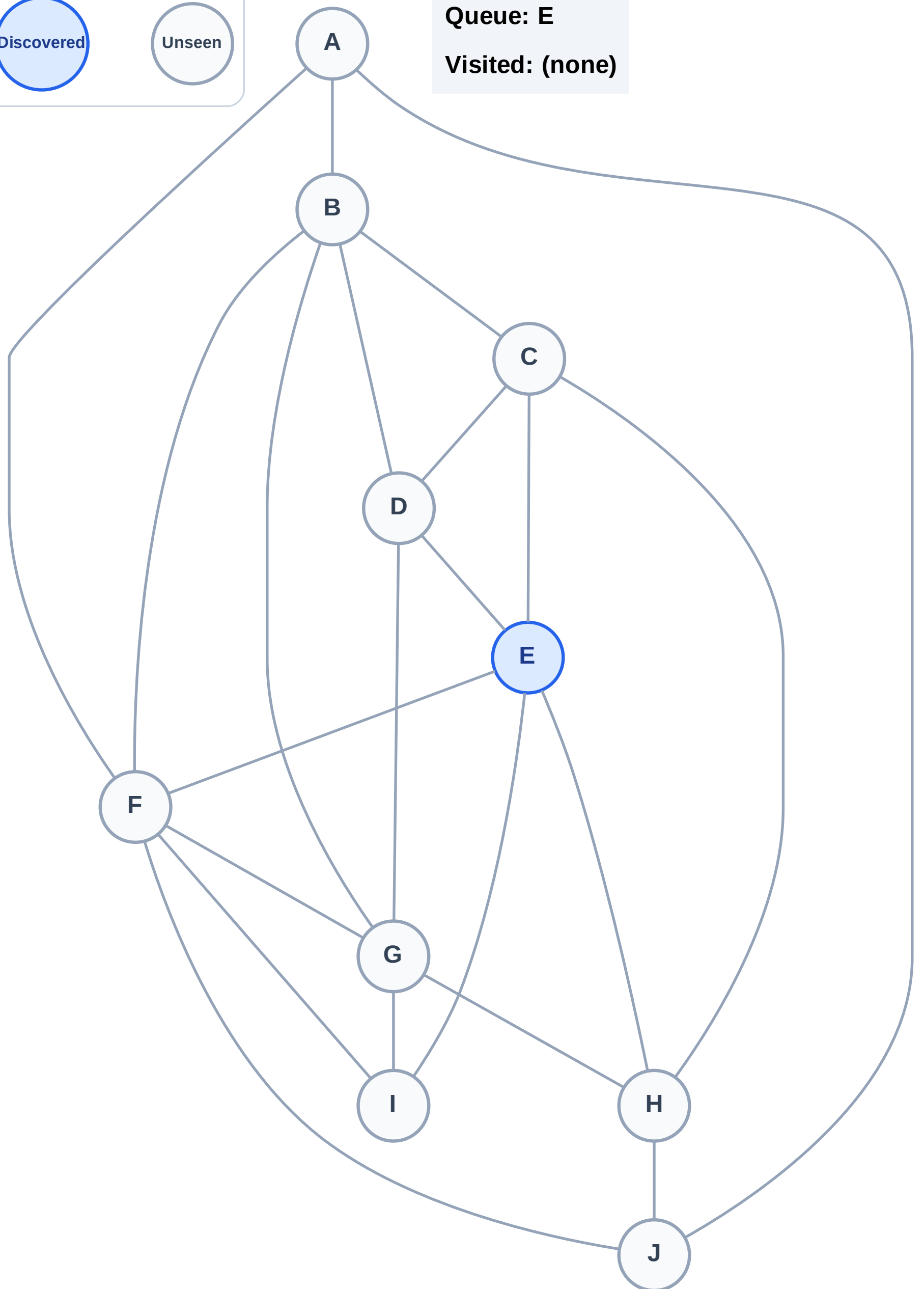
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

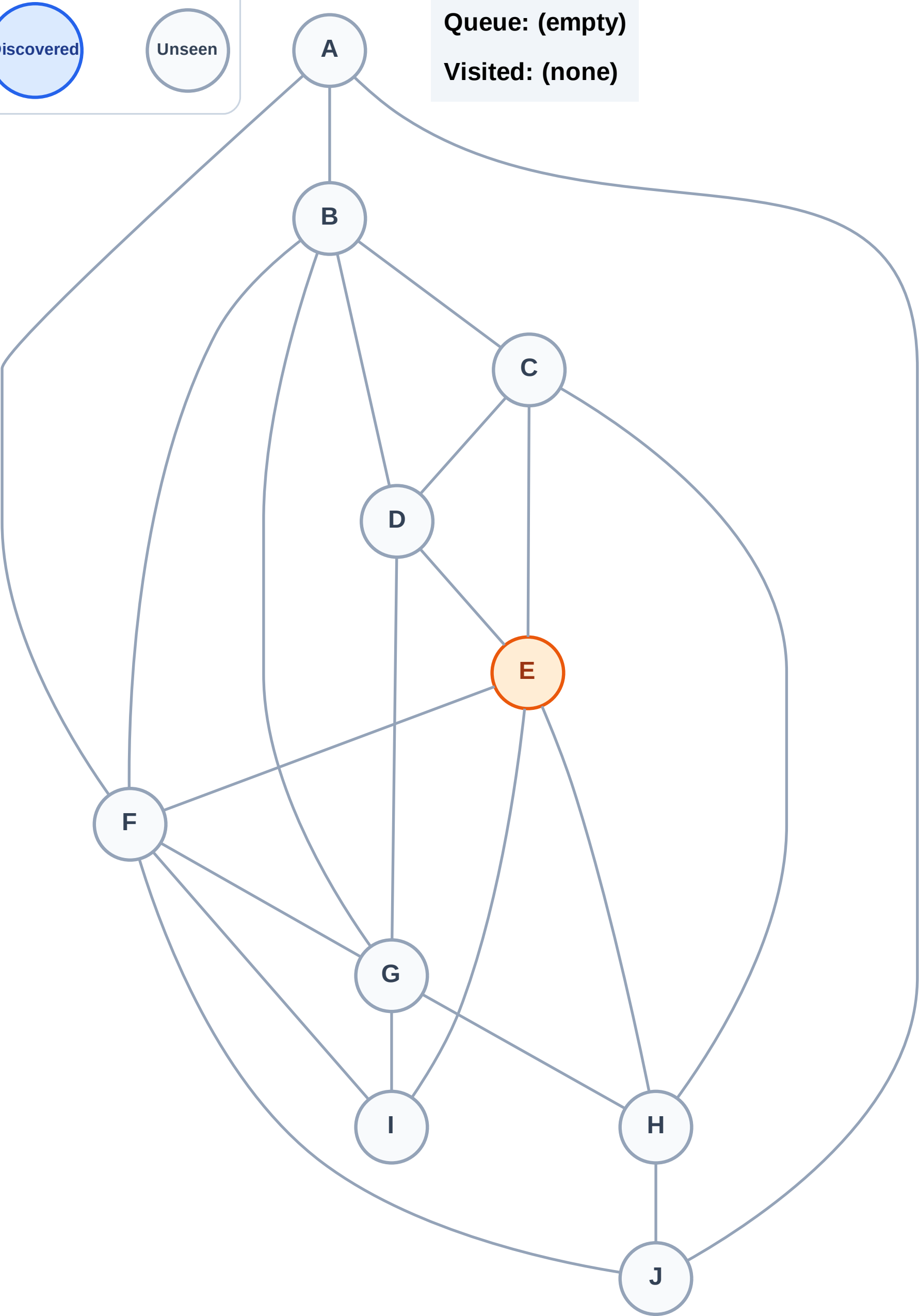
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover C, D, F, H, I from E

Legend

Complete

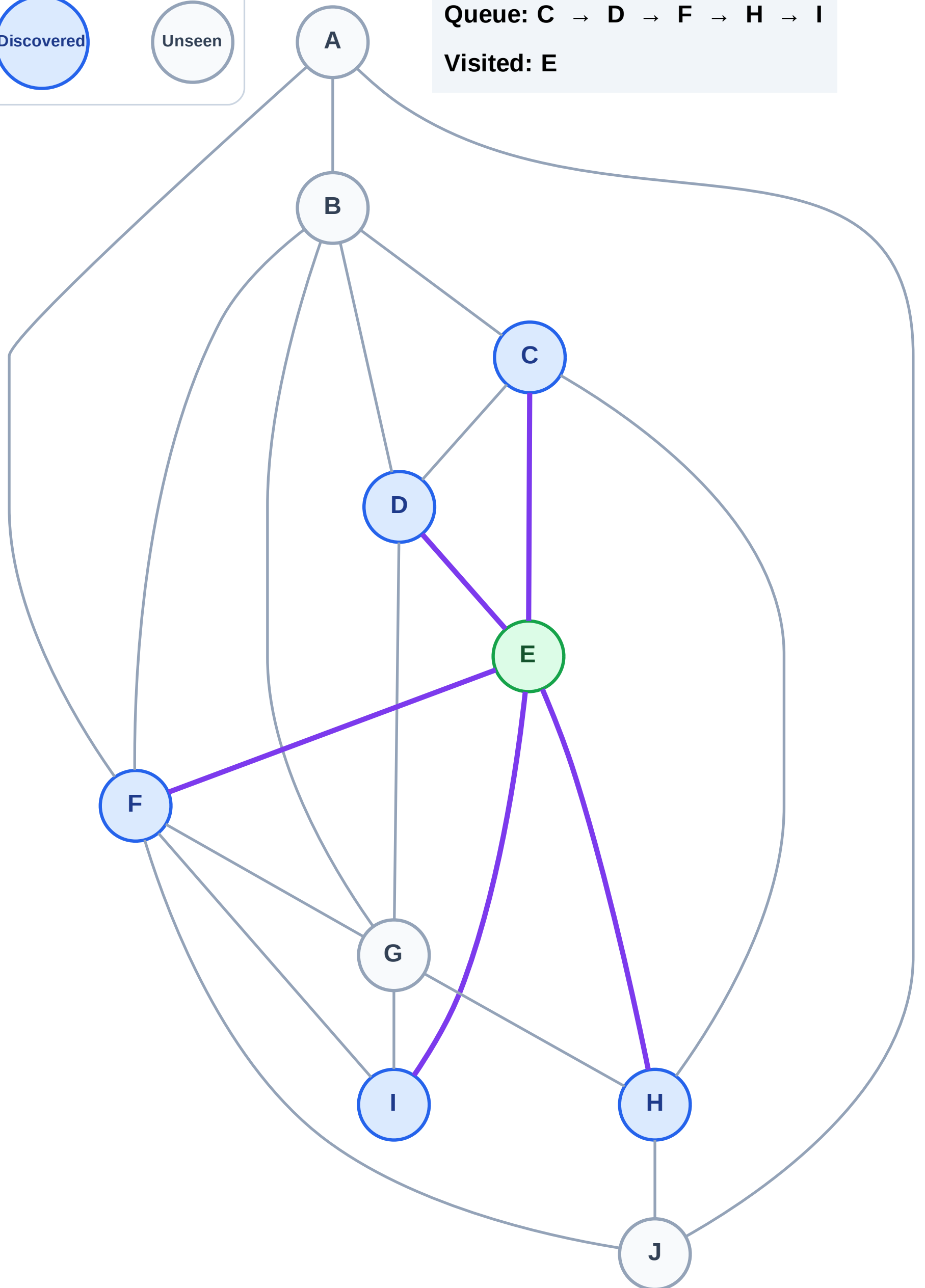
Processing

Discovered

Unseen

Queue: C → D → F → H → I

Visited: E



BFS Traversal

Step 4: Process node C

Legend

Complete

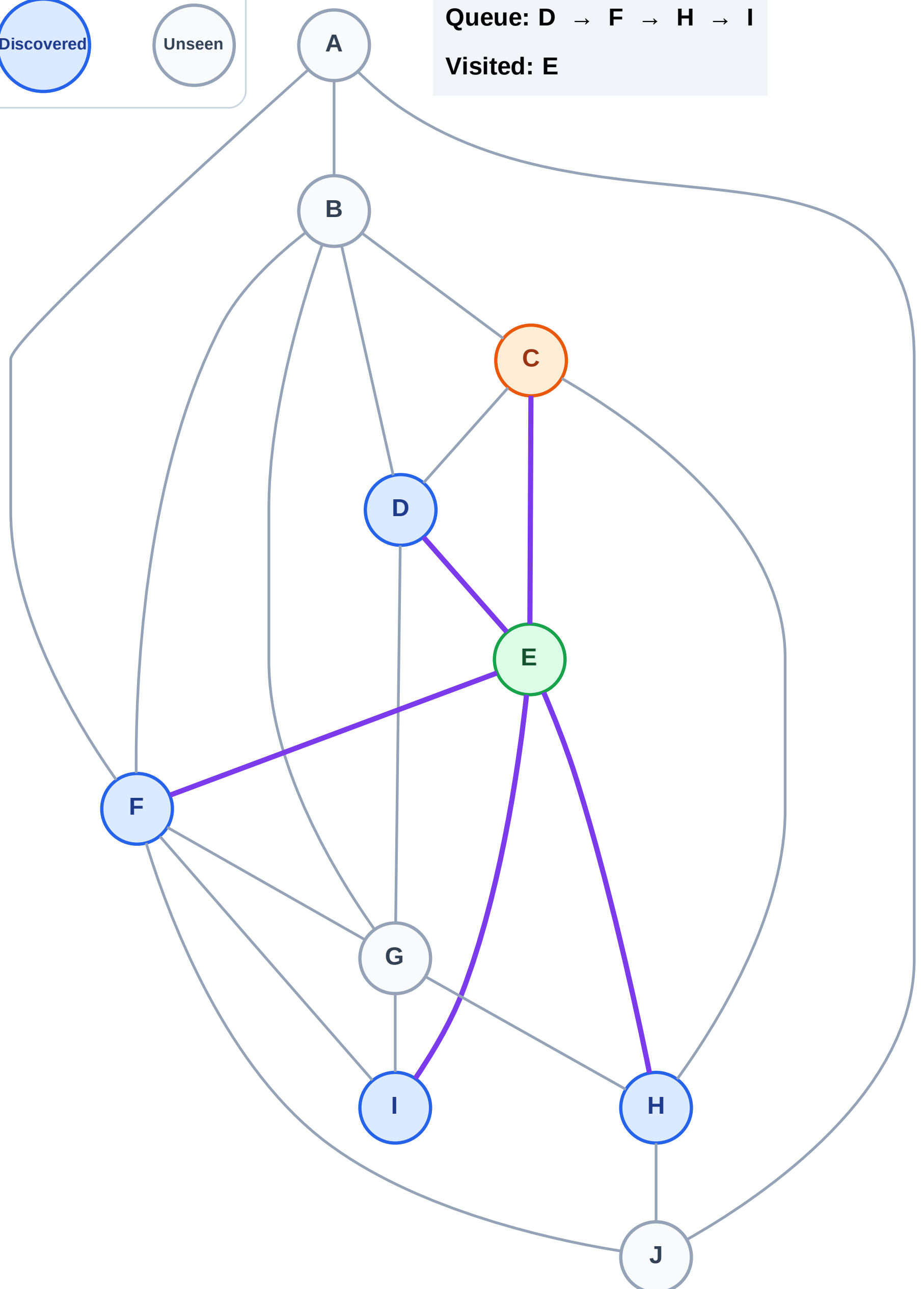
Processing

Discovered

Unseen

Queue: D → F → H → I

Visited: E



BFS Traversal

Step 5: Discover B from C

Legend

Complete

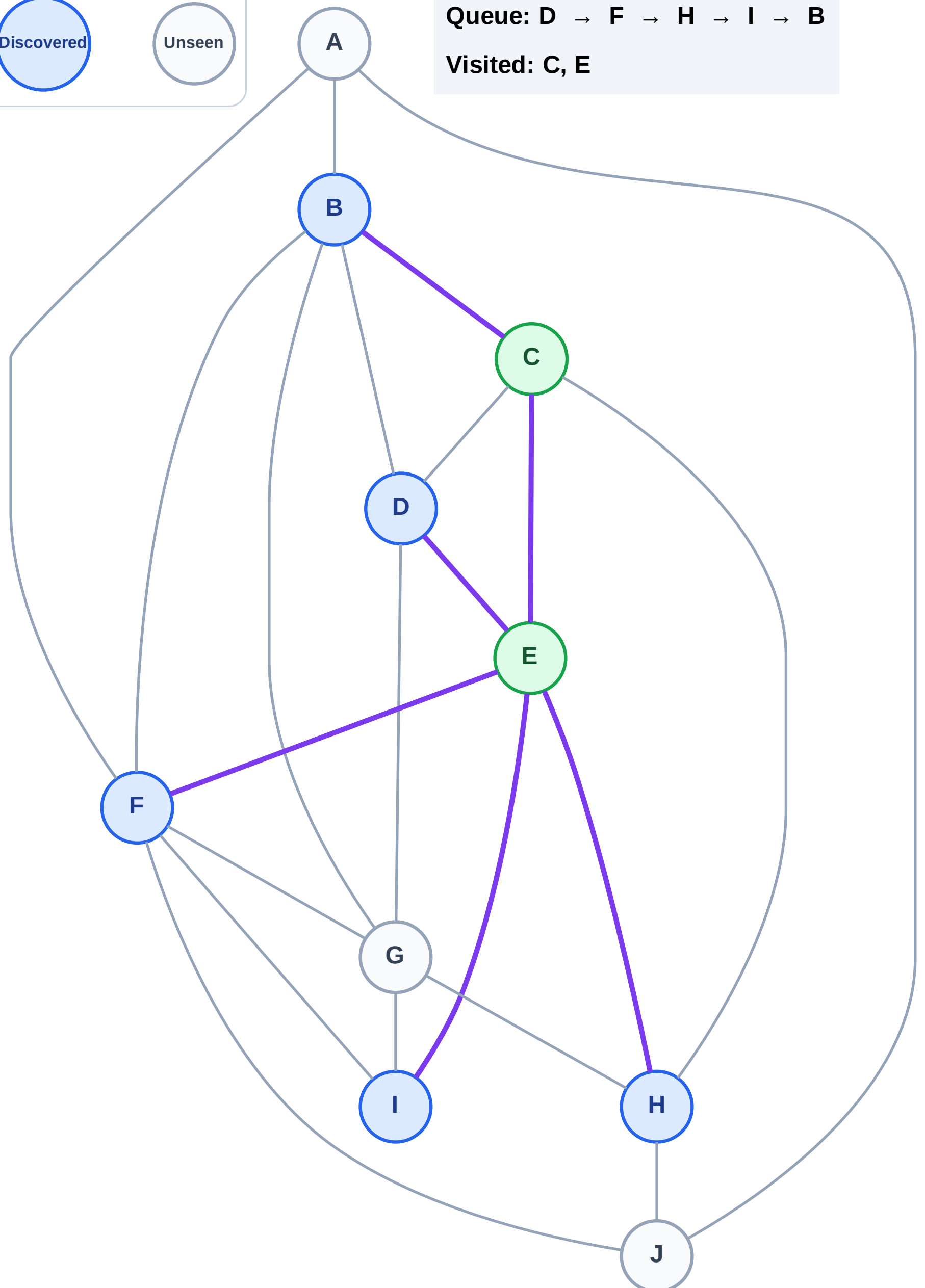
Processing

Discovered

Unseen

Queue: D → F → H → I → B

Visited: C, E



BFS Traversal

Step 6: Process node D

Legend

Complete

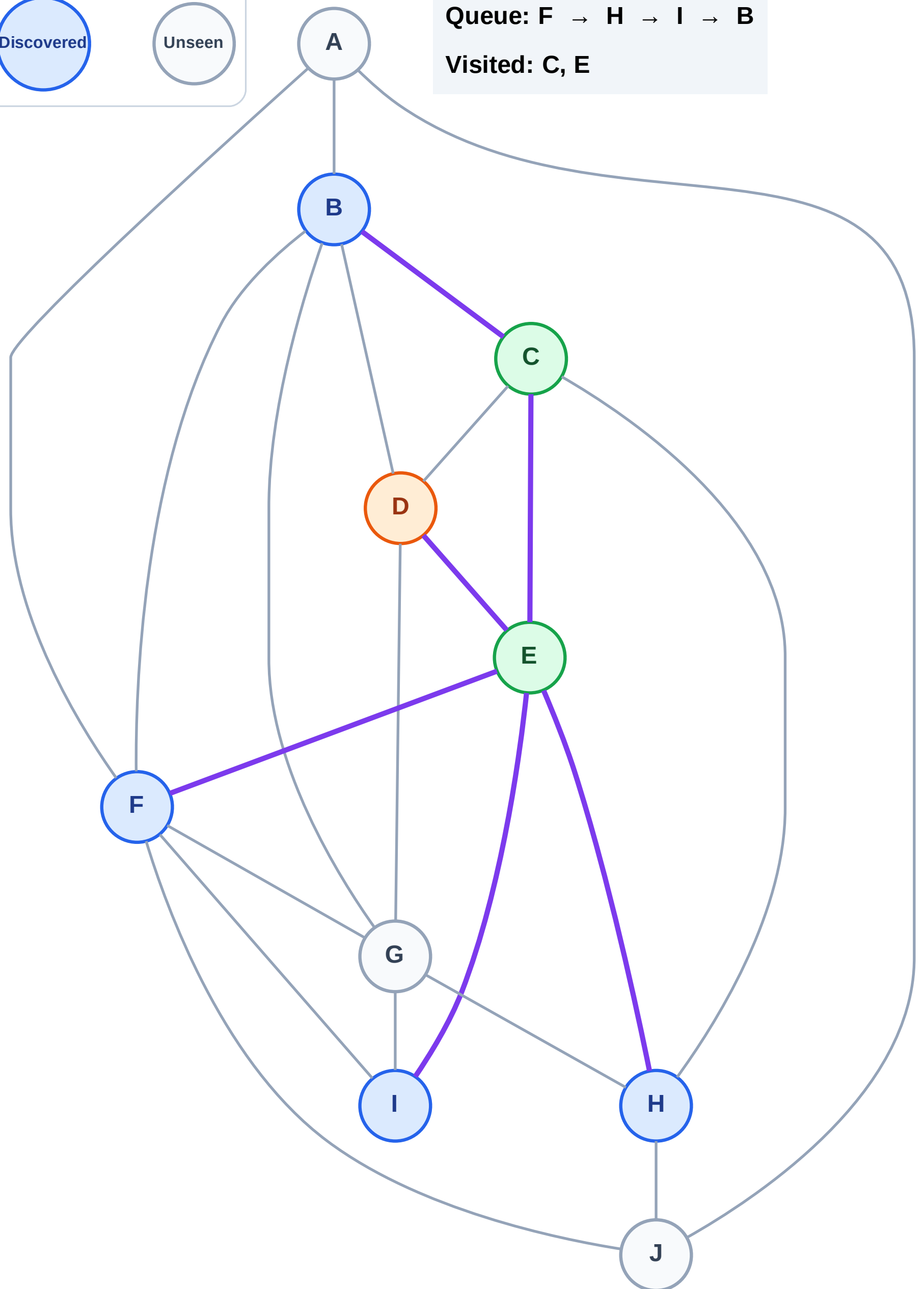
Processing

Discovered

Unseen

Queue: F → H → I → B

Visited: C, E



BFS Traversal

Step 7: Discover G from D

Legend

Complete

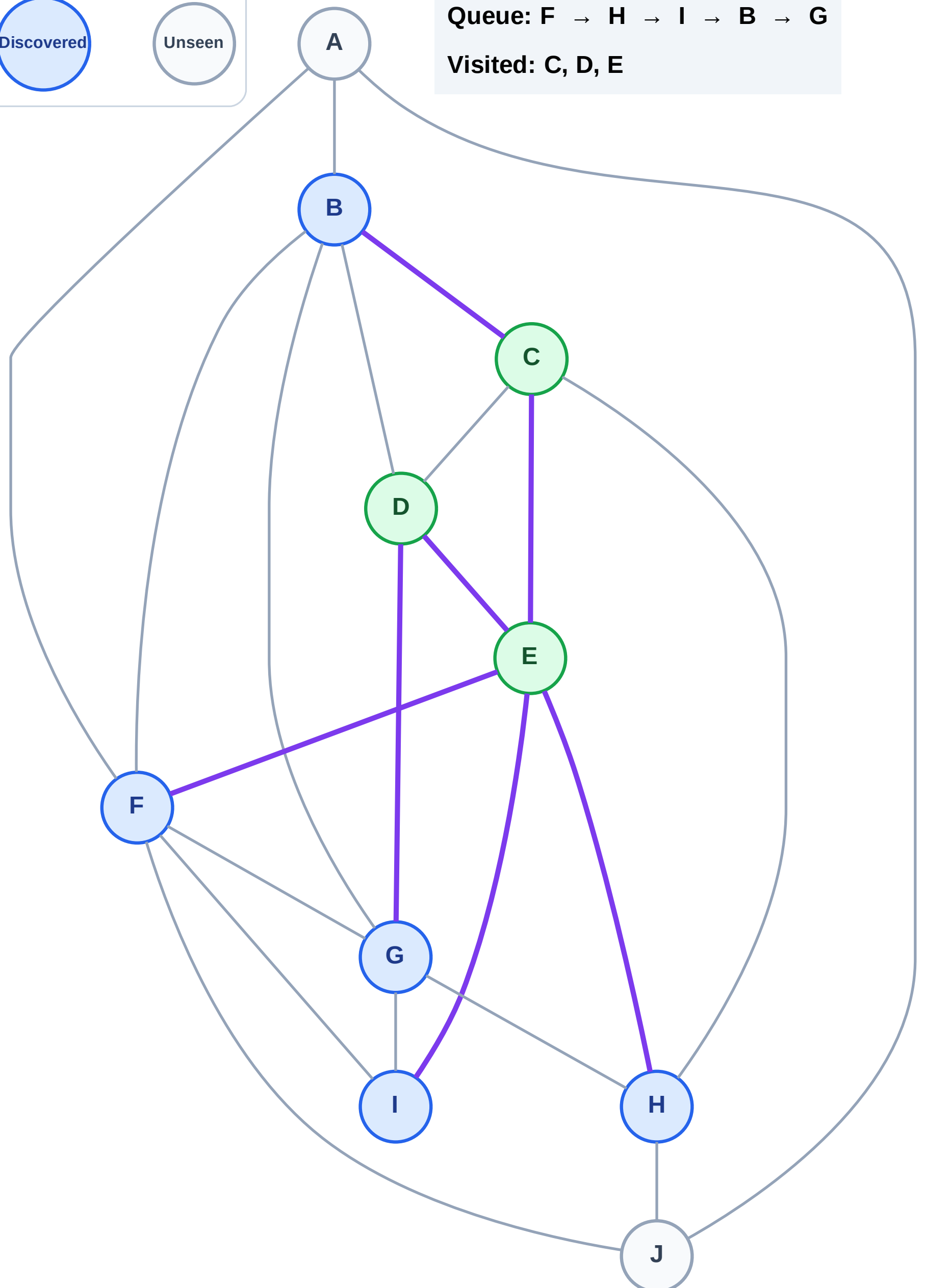
Processing

Discovered

Unseen

Queue: F → H → I → B → G

Visited: C, D, E



BFS Traversal

Step 8: Process node F

Legend

Complete

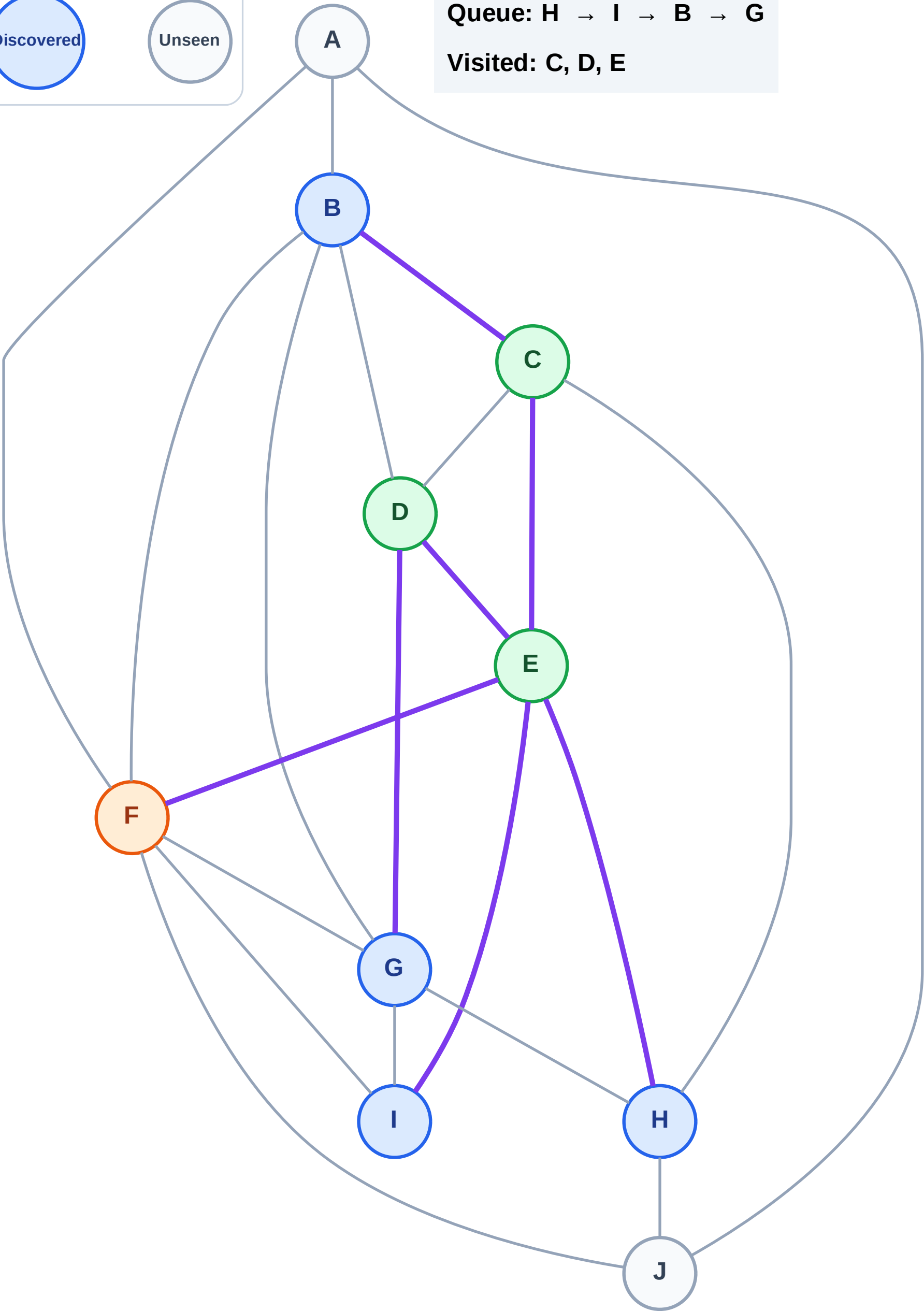
Processing

Discovered

Unseen

Queue: H → I → B → G

Visited: C, D, E



BFS Traversal

Step 9: Discover A, J from F

Legend

Complete

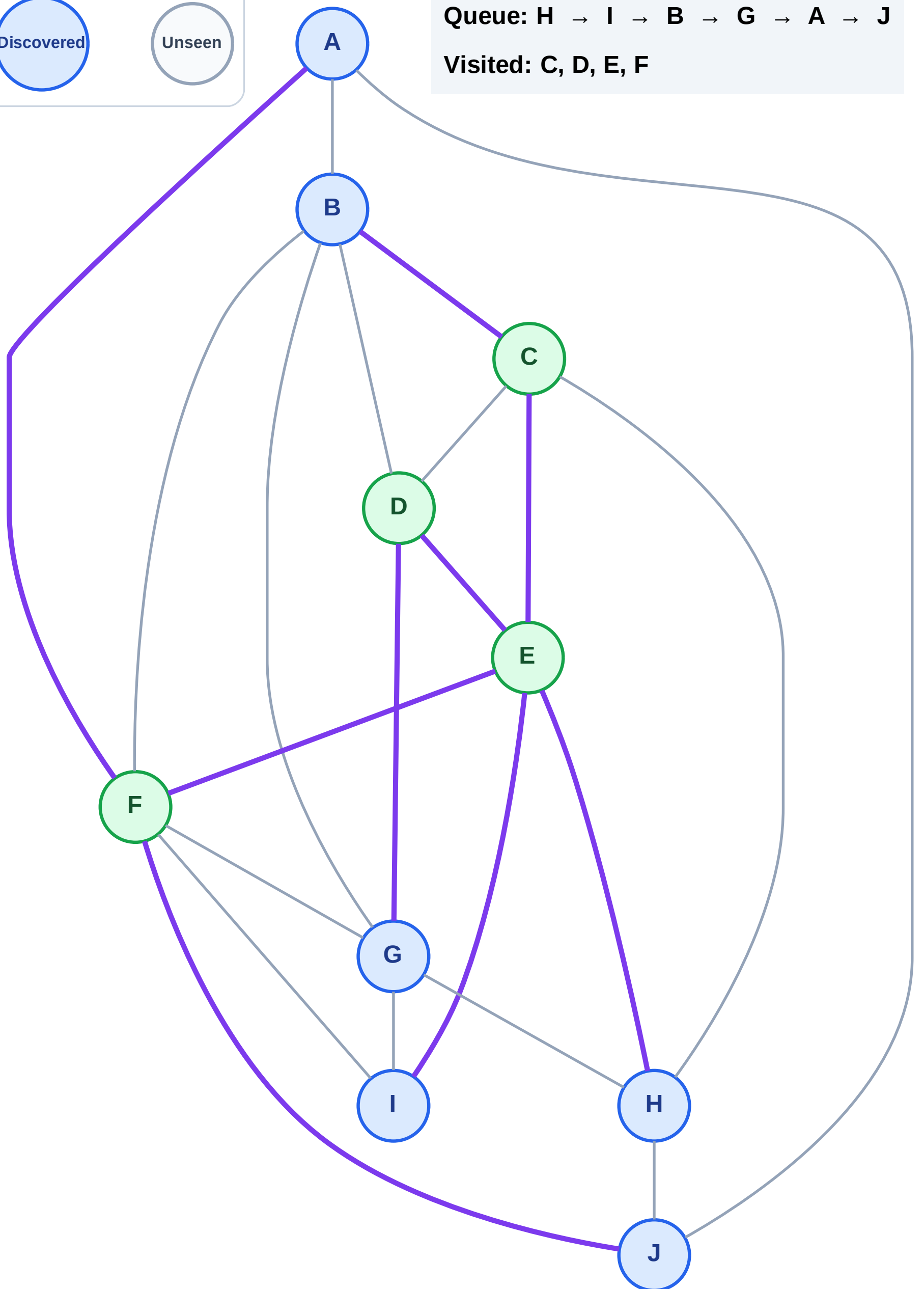
Processing

Discovered

Unseen

Queue: H → I → B → G → A → J

Visited: C, D, E, F



BFS Traversal

Step 10: Process node H

Legend

Complete

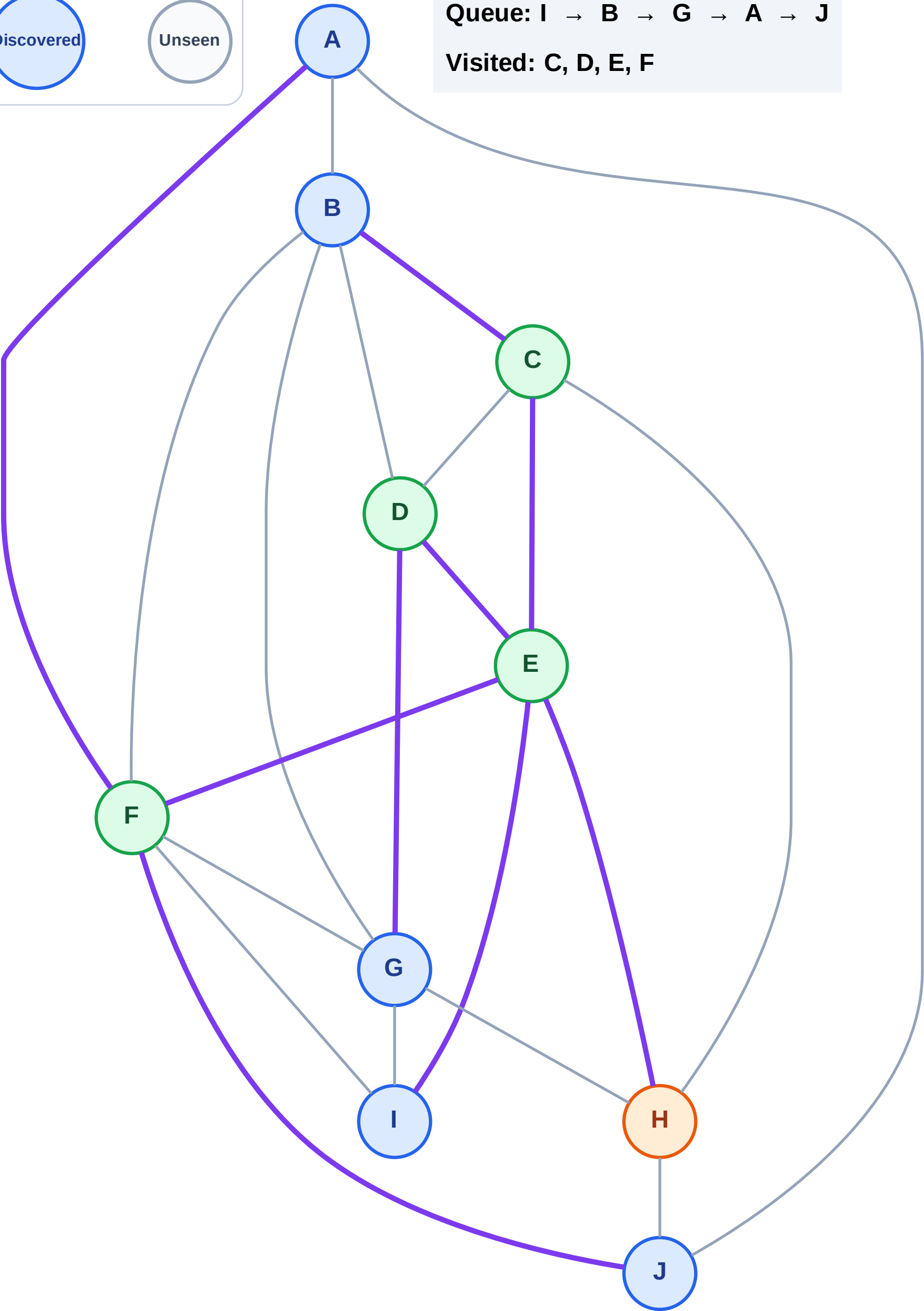
Processing

Discovered

Unseen

Queue: I → B → G → A → J

Visited: C, D, E, F



BFS Traversal

Step 11: No new neighbors from H

Legend

Complete

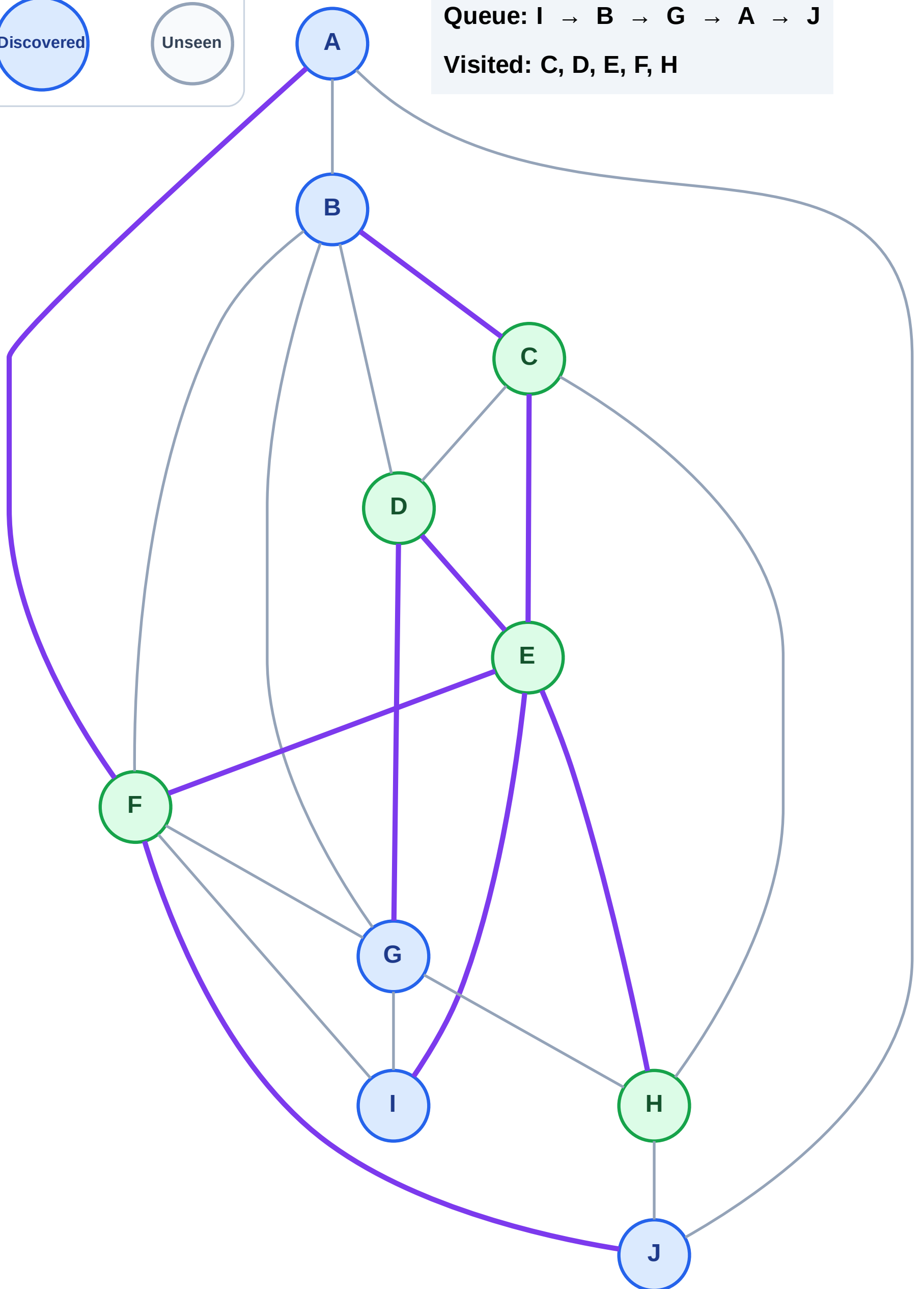
Processing

Discovered

Unseen

Queue: I → B → G → A → J

Visited: C, D, E, F, H



BFS Traversal

Step 12: Process node I

Legend

Complete

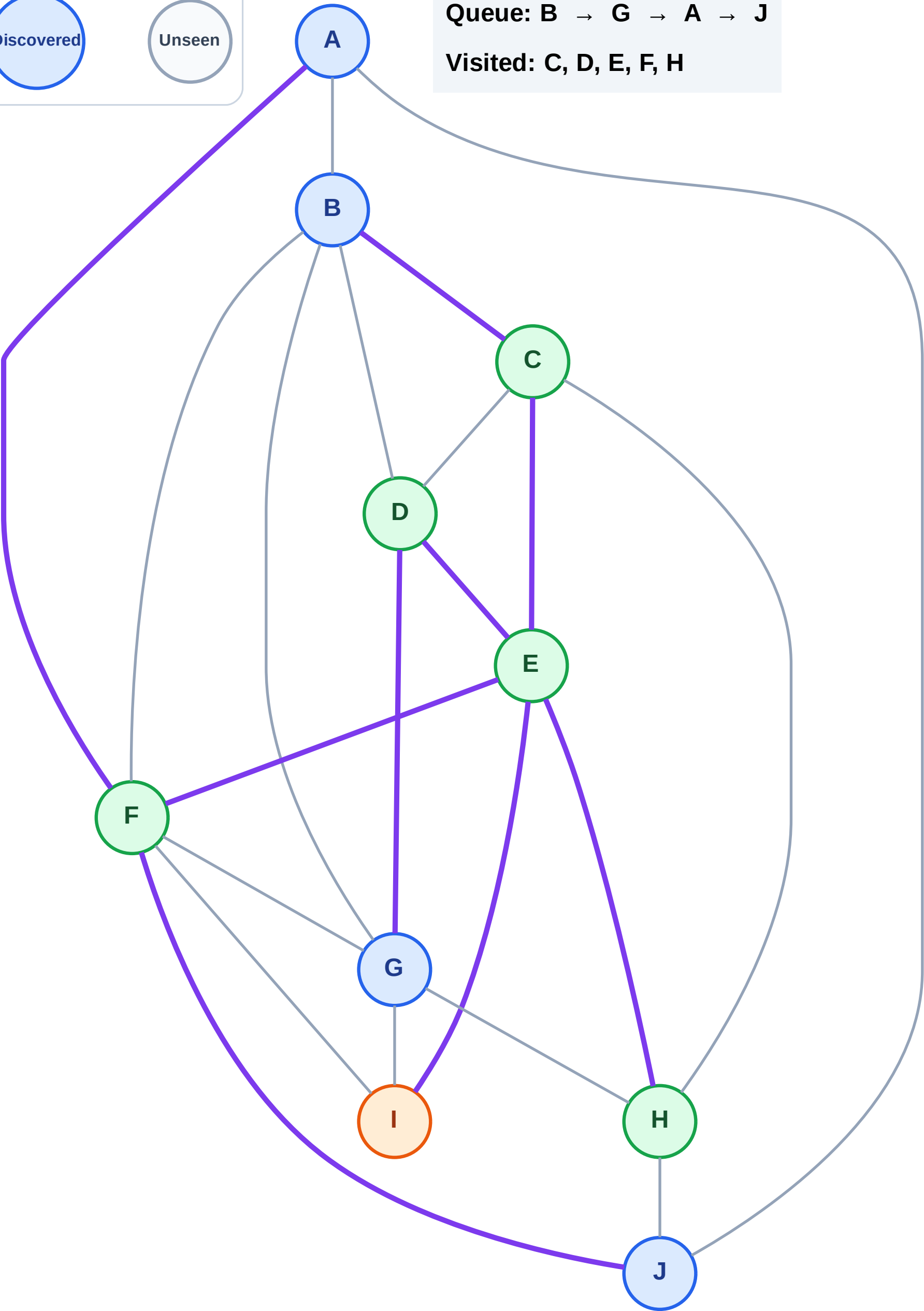
Processing

Discovered

Unseen

Queue: B → G → A → J

Visited: C, D, E, F, H



BFS Traversal

Step 13: No new neighbors from I

Legend

Complete

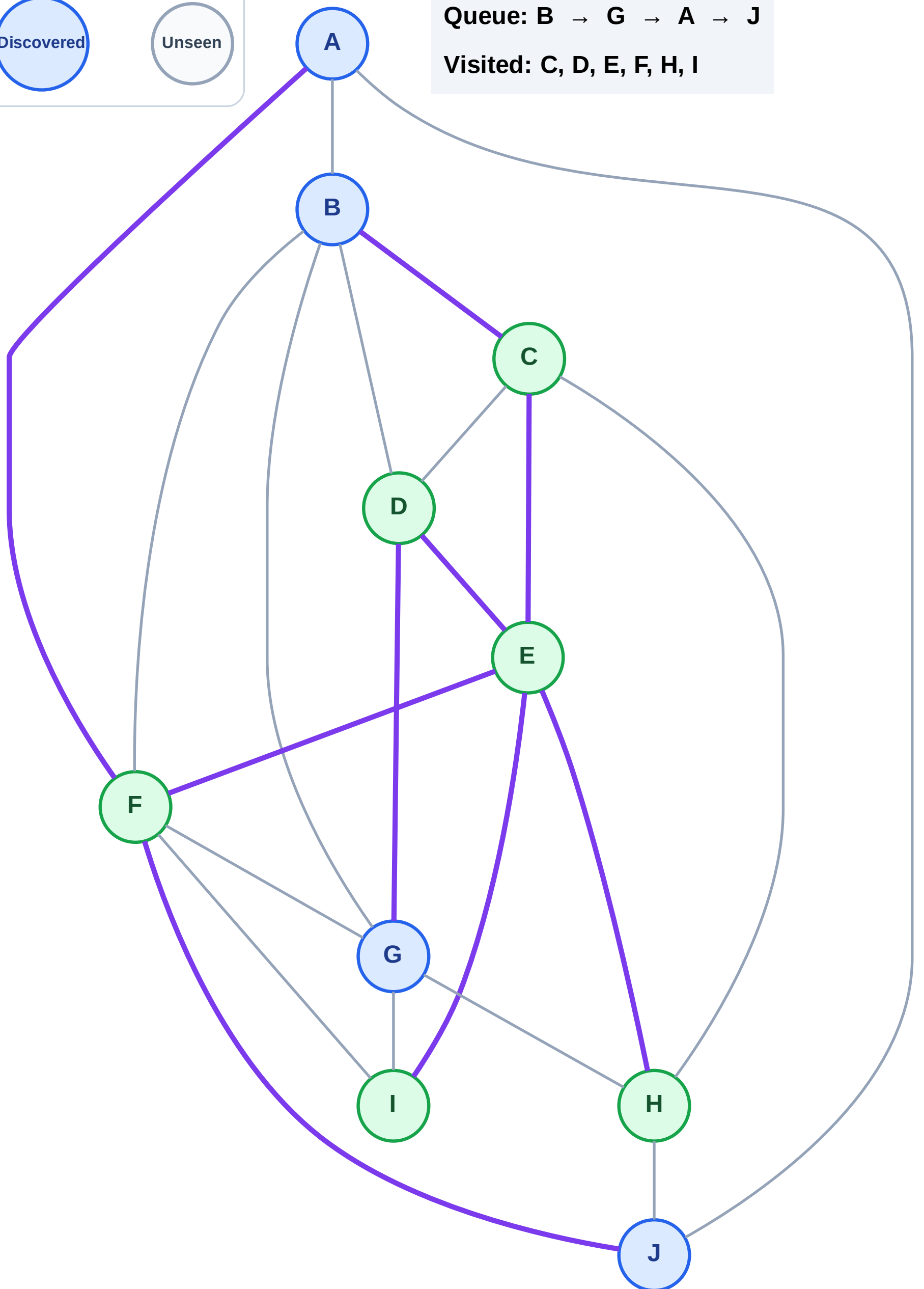
Processing

Discovered

Unseen

Queue: B → G → A → J

Visited: C, D, E, F, H, I



BFS Traversal

Step 14: Process node B

Legend

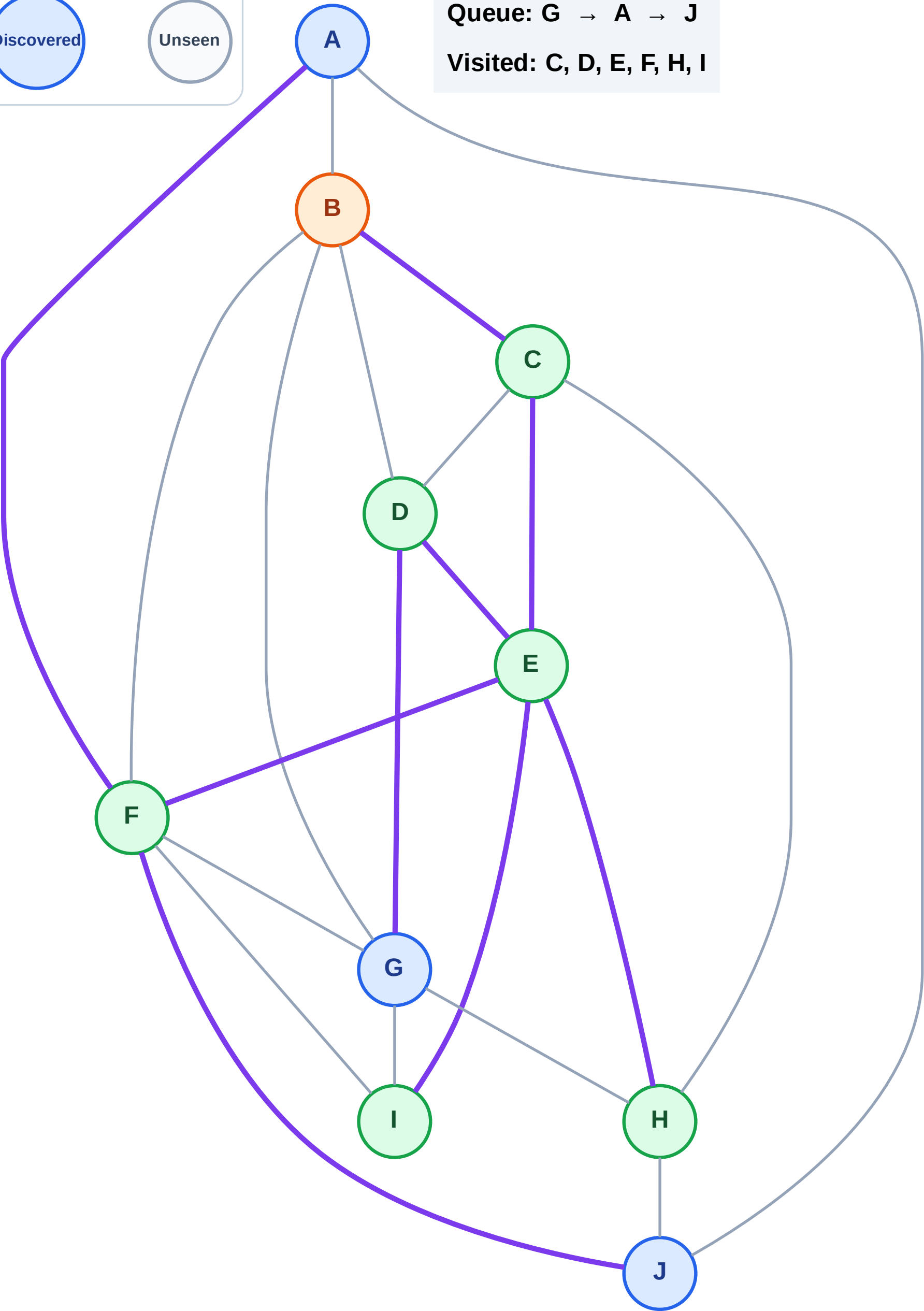
Complete

Processing

Discovered

Unseen

Queue: G → A → J
Visited: C, D, E, F, H, I



BFS Traversal

Step 15: No new neighbors from B

Legend

Complete

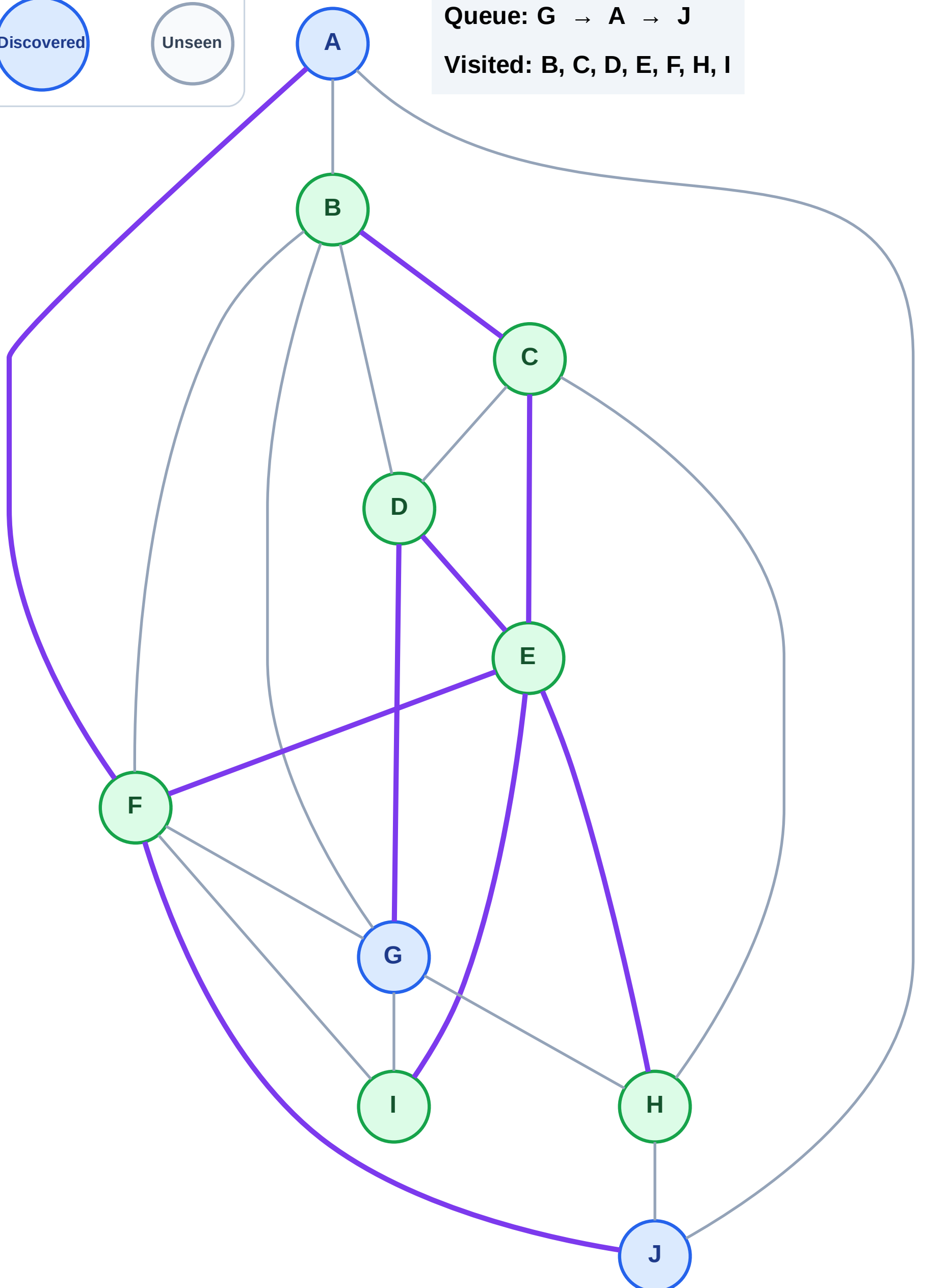
Processing

Discovered

Unseen

Queue: G → A → J

Visited: B, C, D, E, F, H, I



BFS Traversal

Step 16: Process node G

Legend

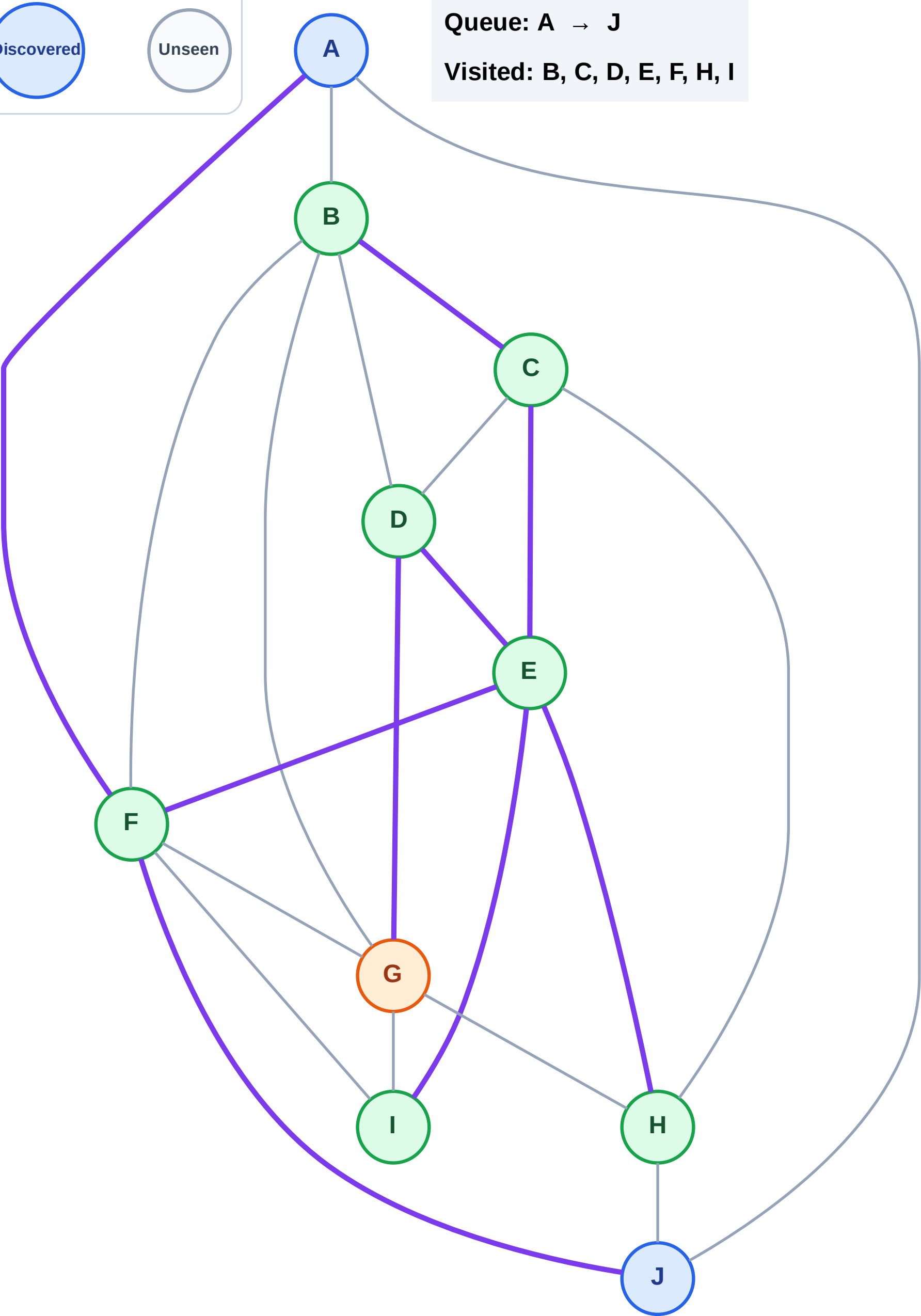
Complete

Processing

Discovered

Unseen

Queue: A → J
Visited: B, C, D, E, F, H, I



BFS Traversal

Step 17: No new neighbors from G

Legend

Complete

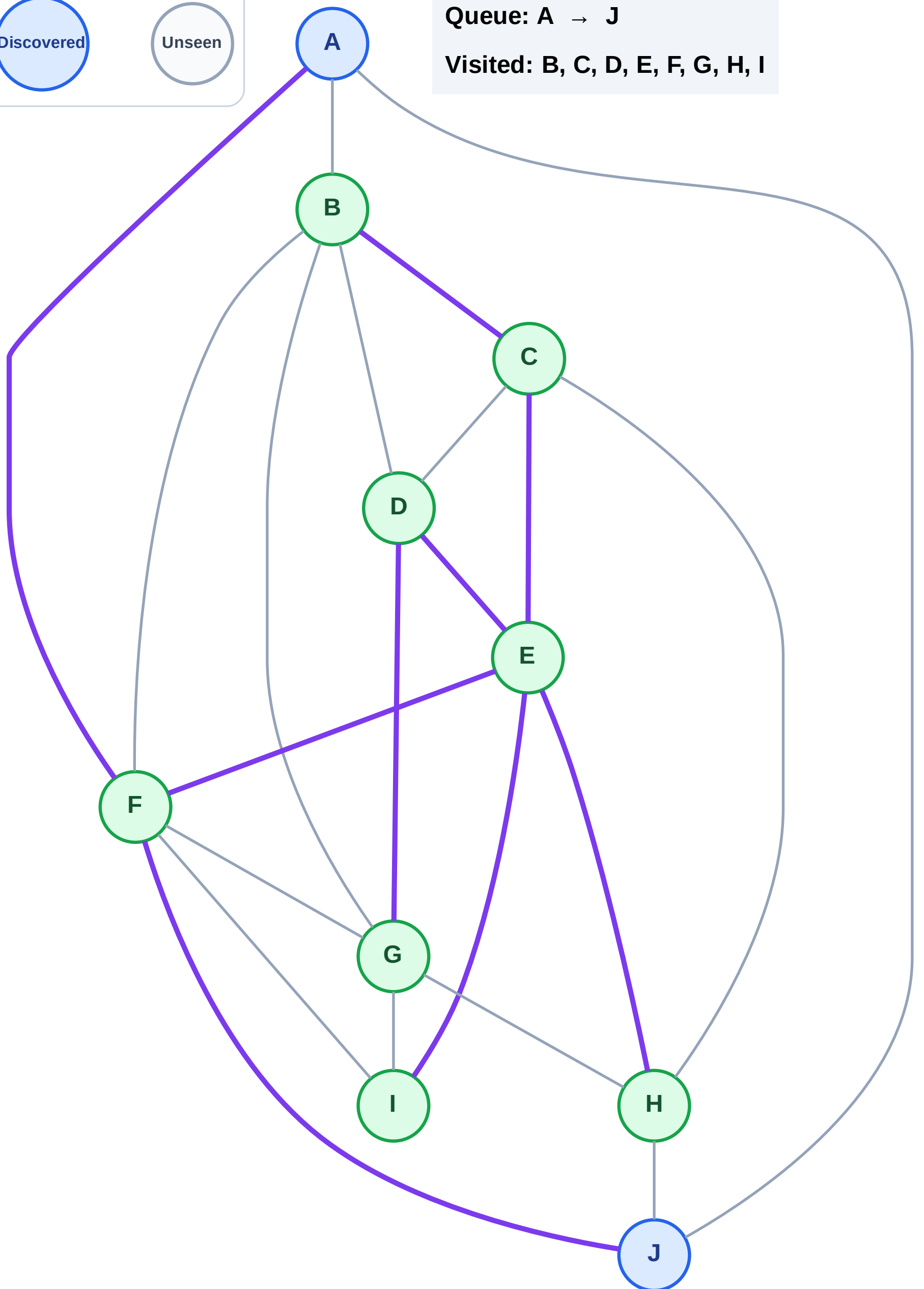
Processing

Discovered

Unseen

Queue: A → J

Visited: B, C, D, E, F, G, H, I



BFS Traversal

Step 18: Process node A

Legend

Complete

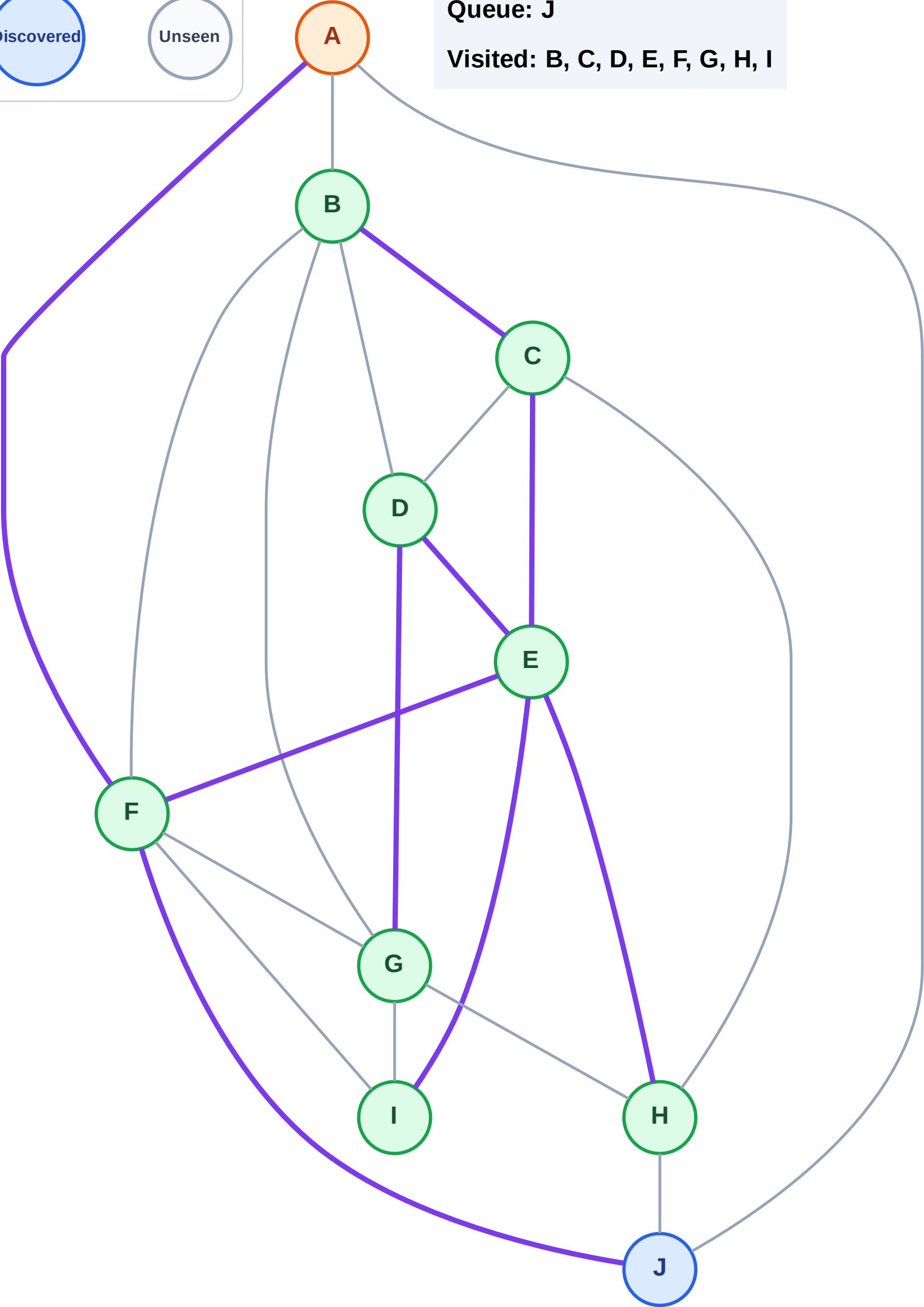
Processing

Discovered

Unseen

Queue: J

Visited: B, C, D, E, F, G, H, I



BFS Traversal

Step 19: No new neighbors from A

Legend

Complete

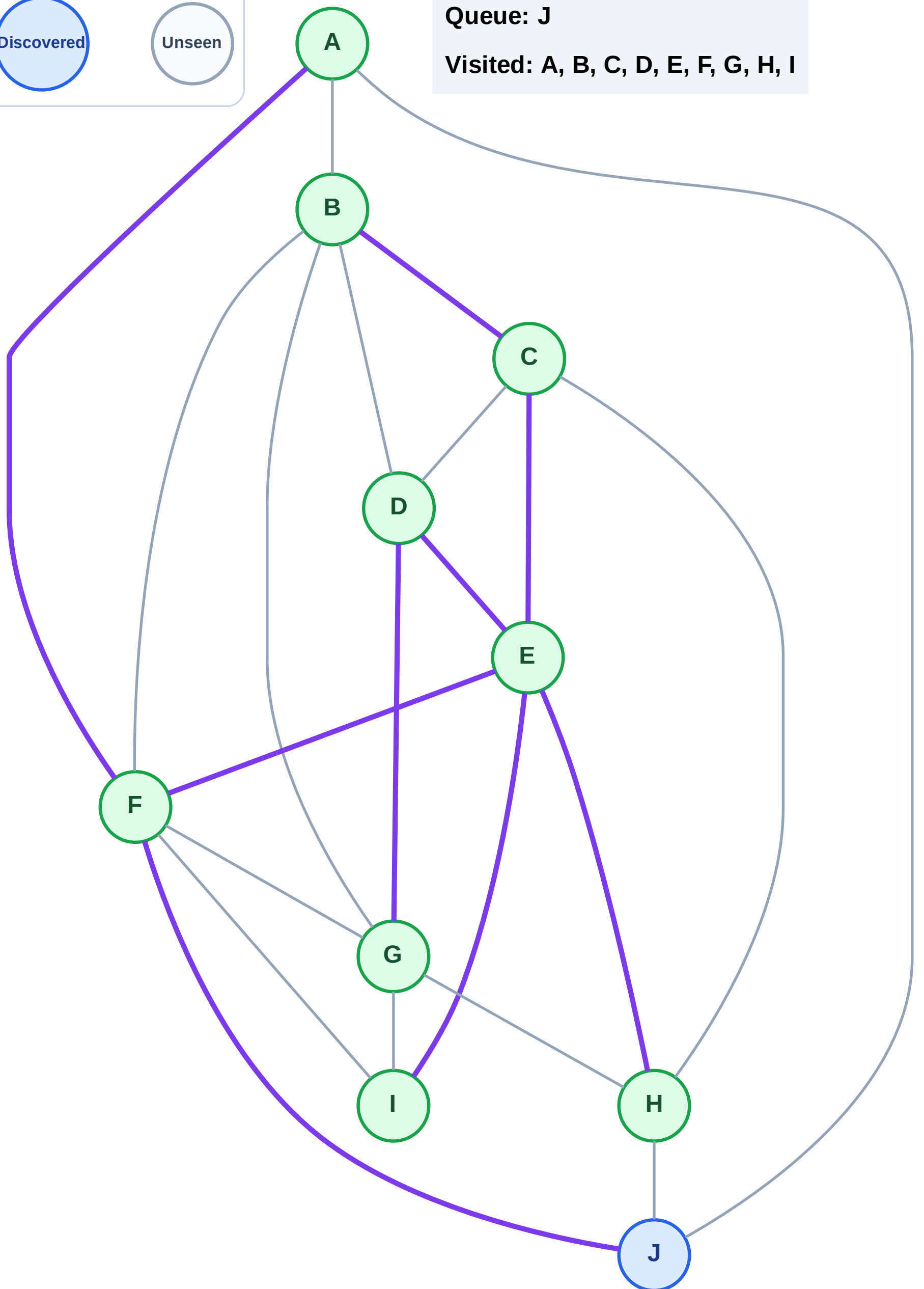
Processing

Discovered

Unseen

Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node J

Legend

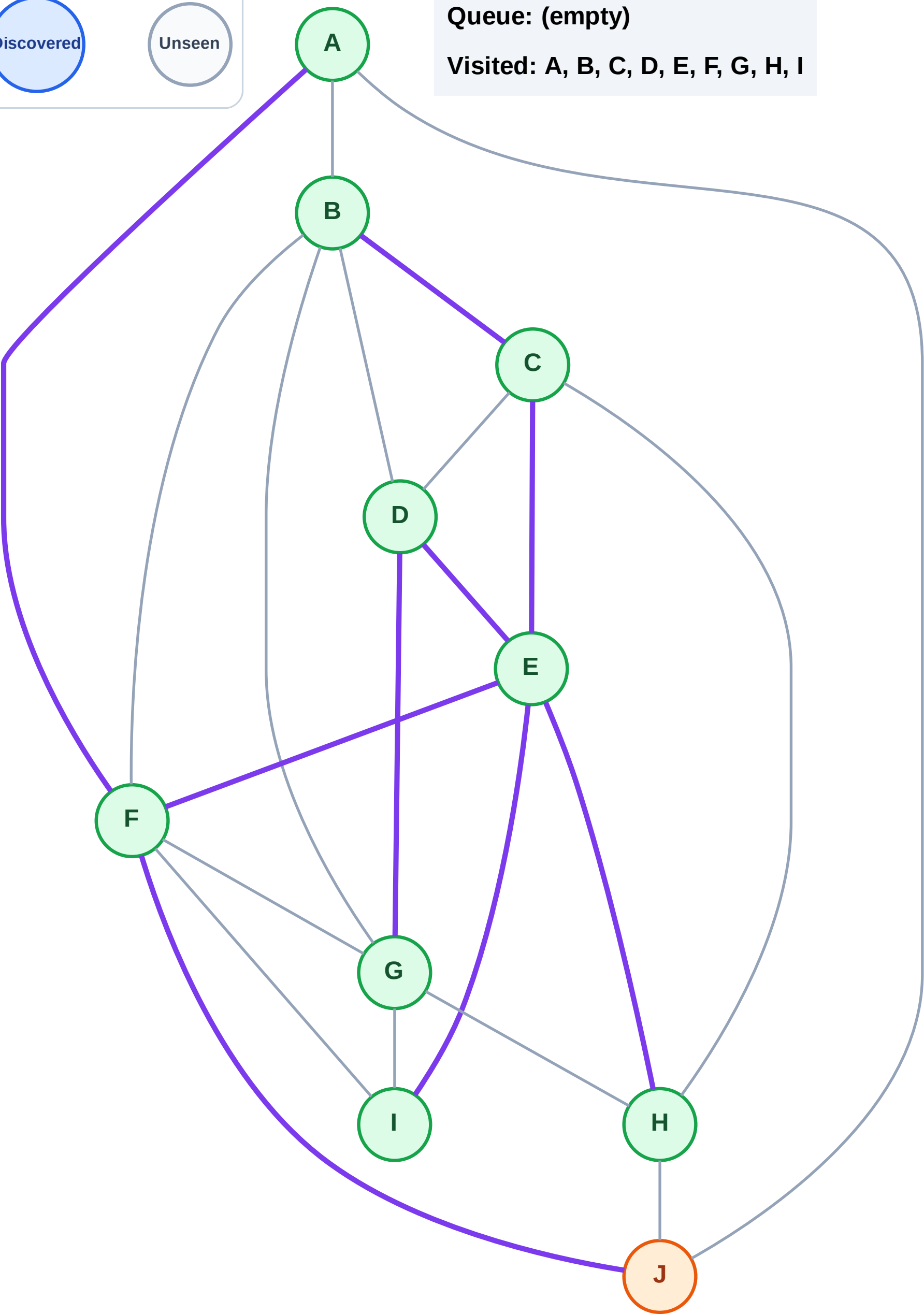
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 21: No new neighbors from J

Legend

Complete

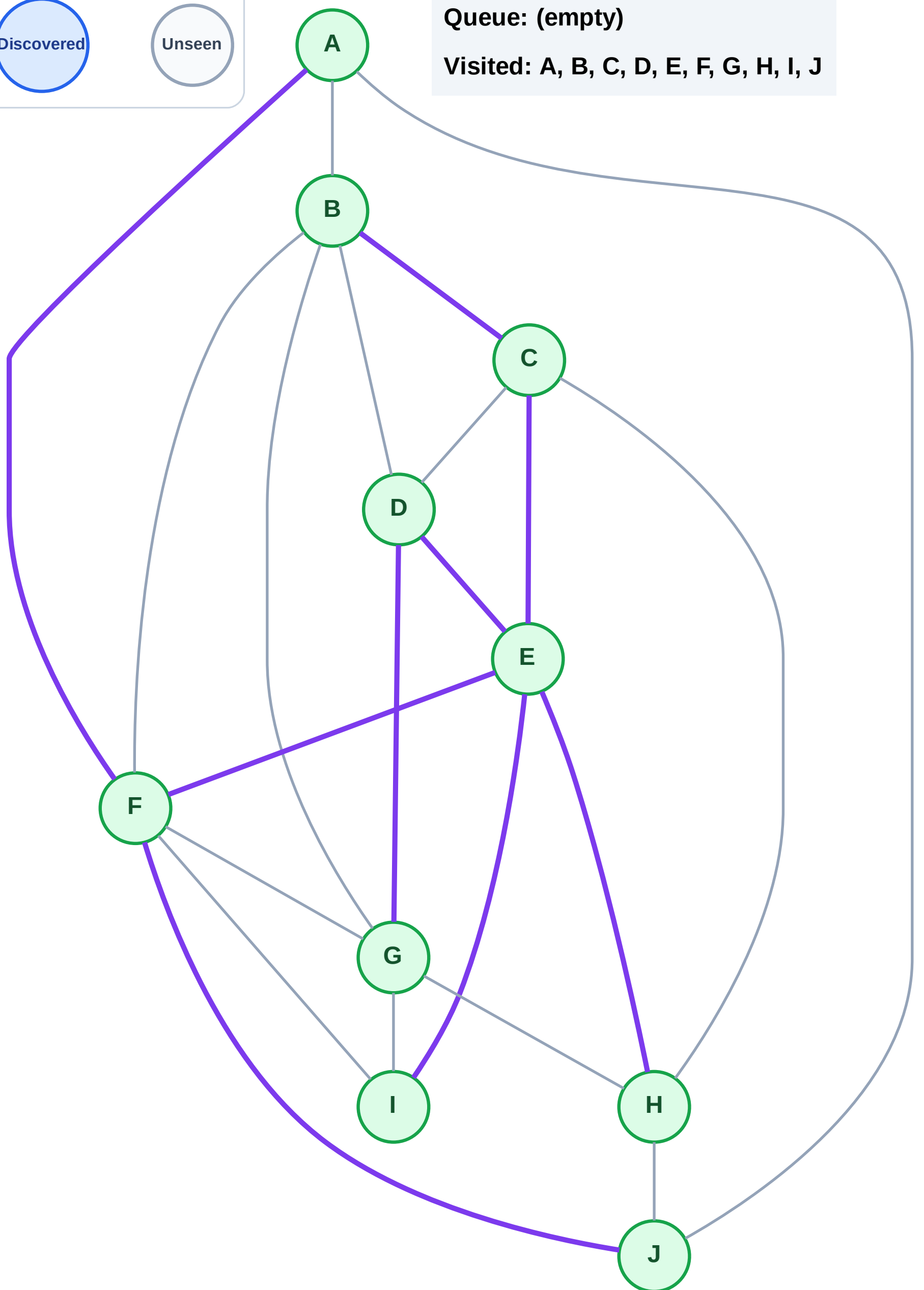
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

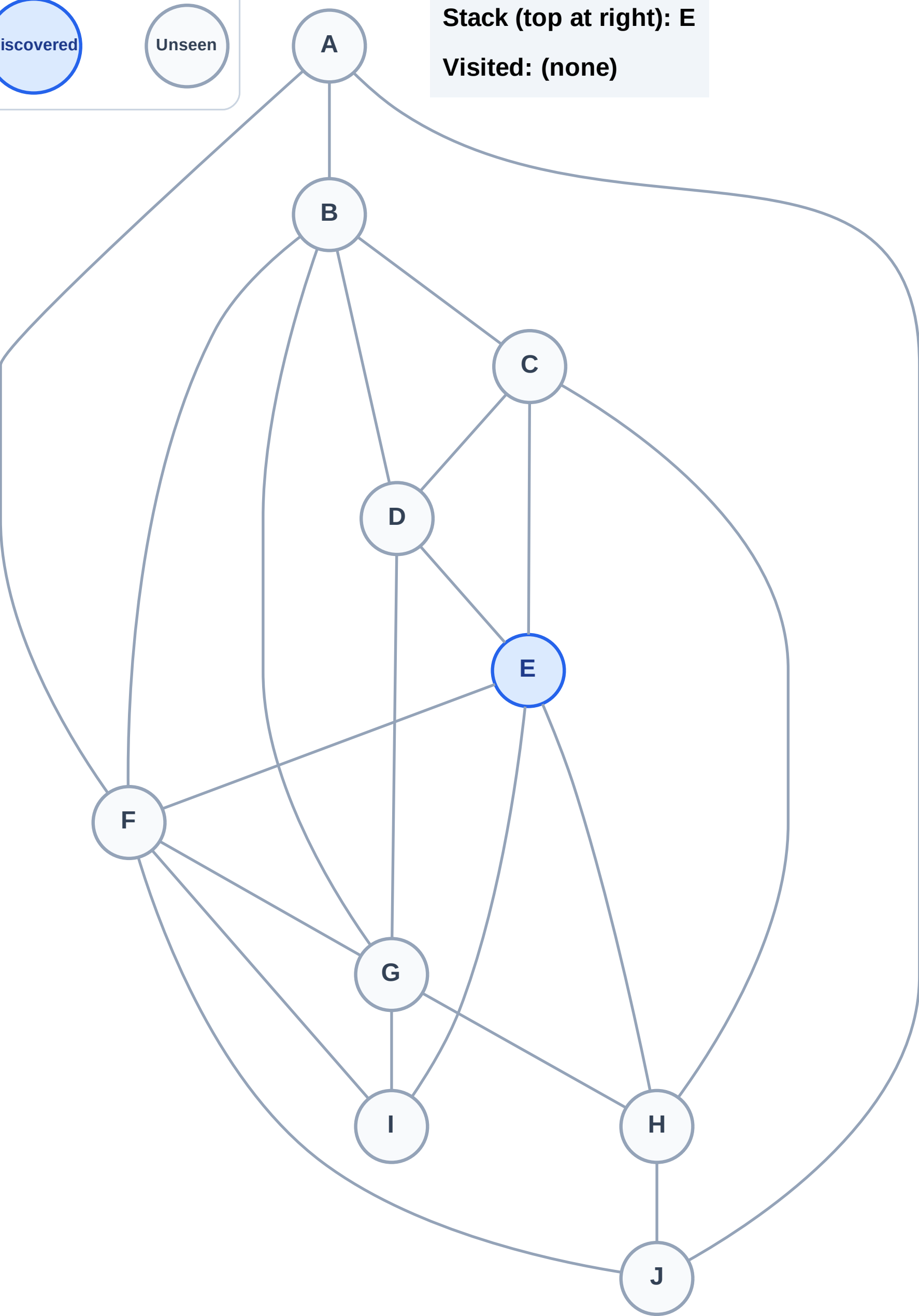
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

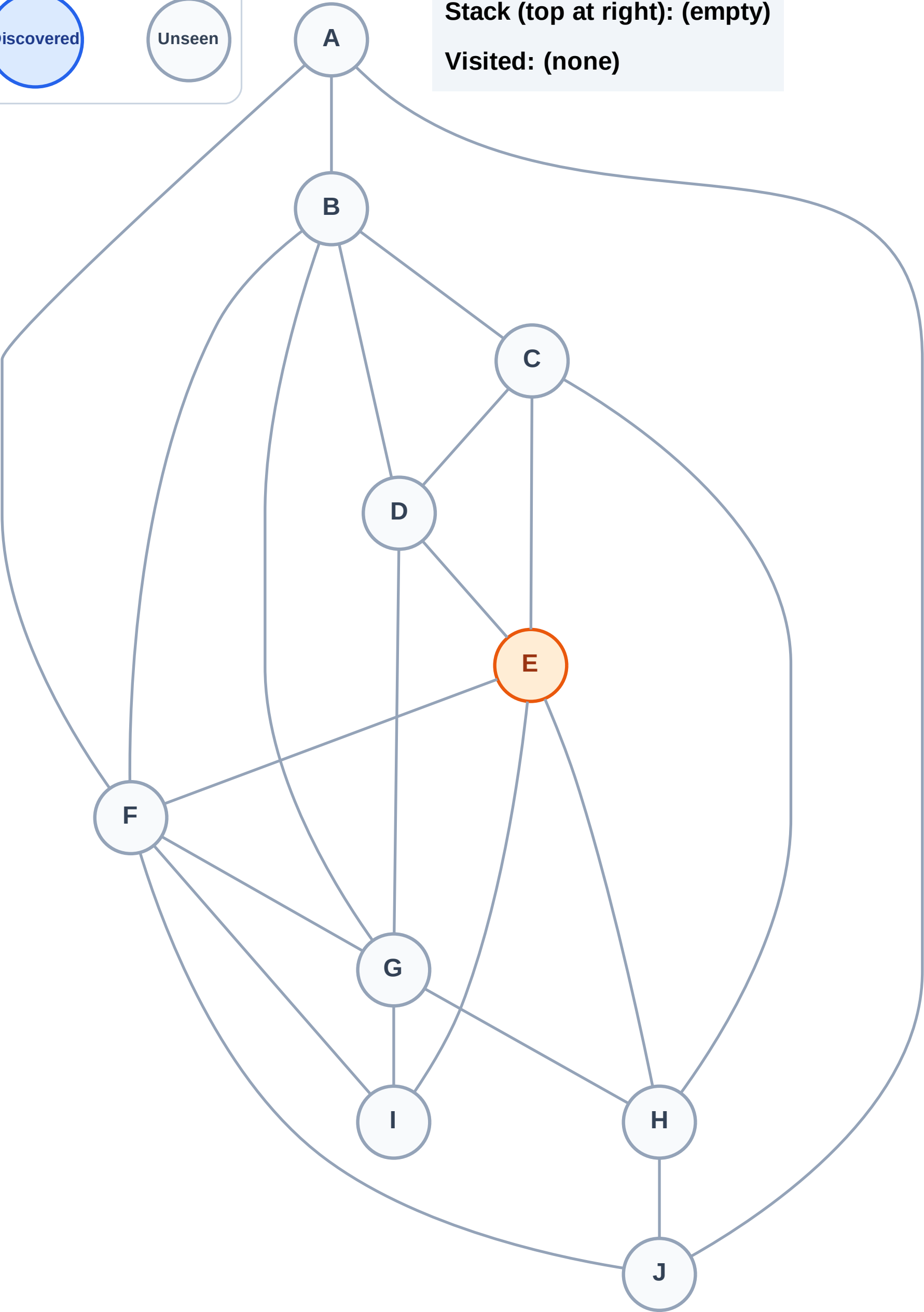
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover I, H, F, D, C from E

Legend

Complete

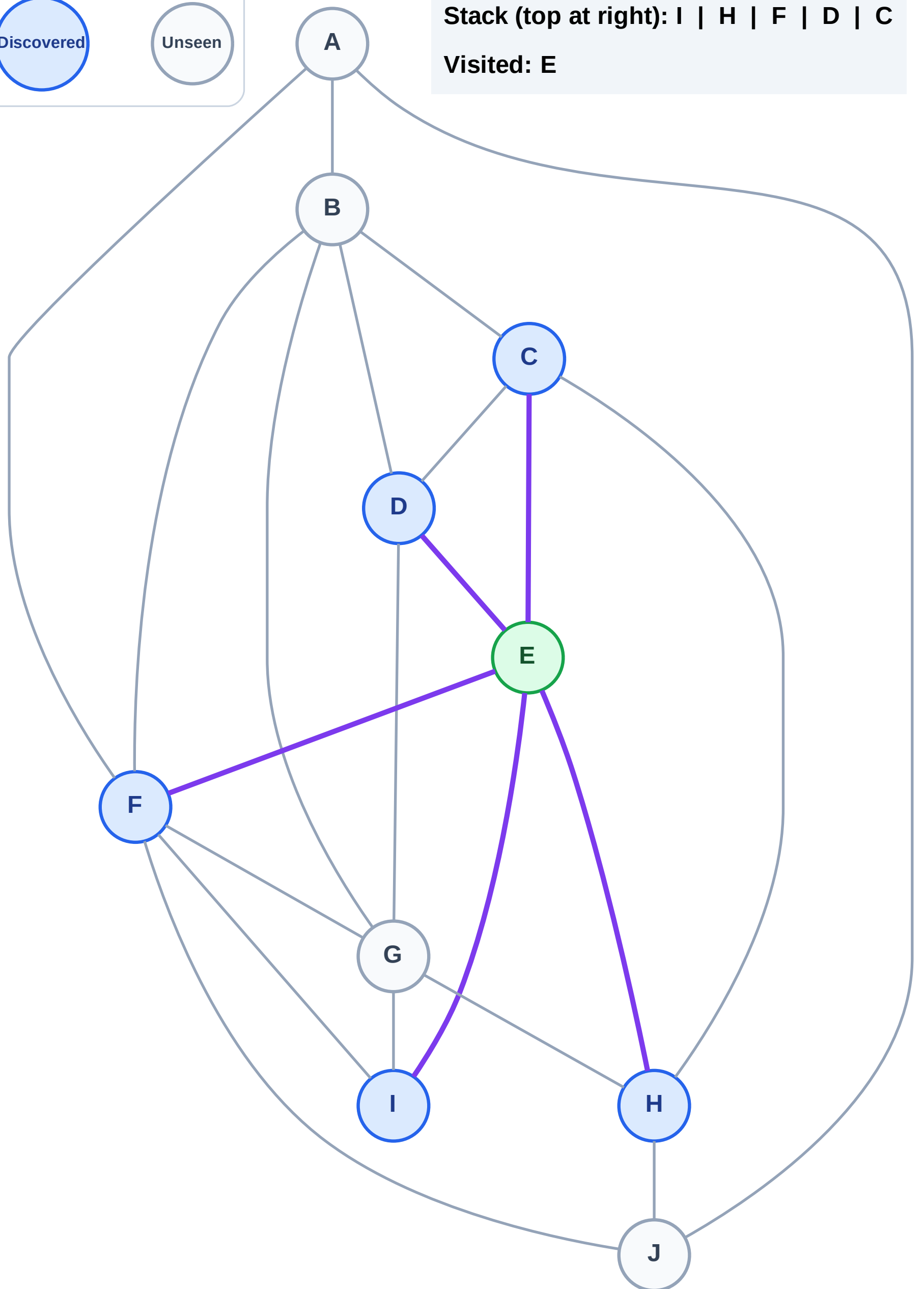
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | C

Visited: E



DFS Traversal

Step 4: Process node C

Legend

Complete

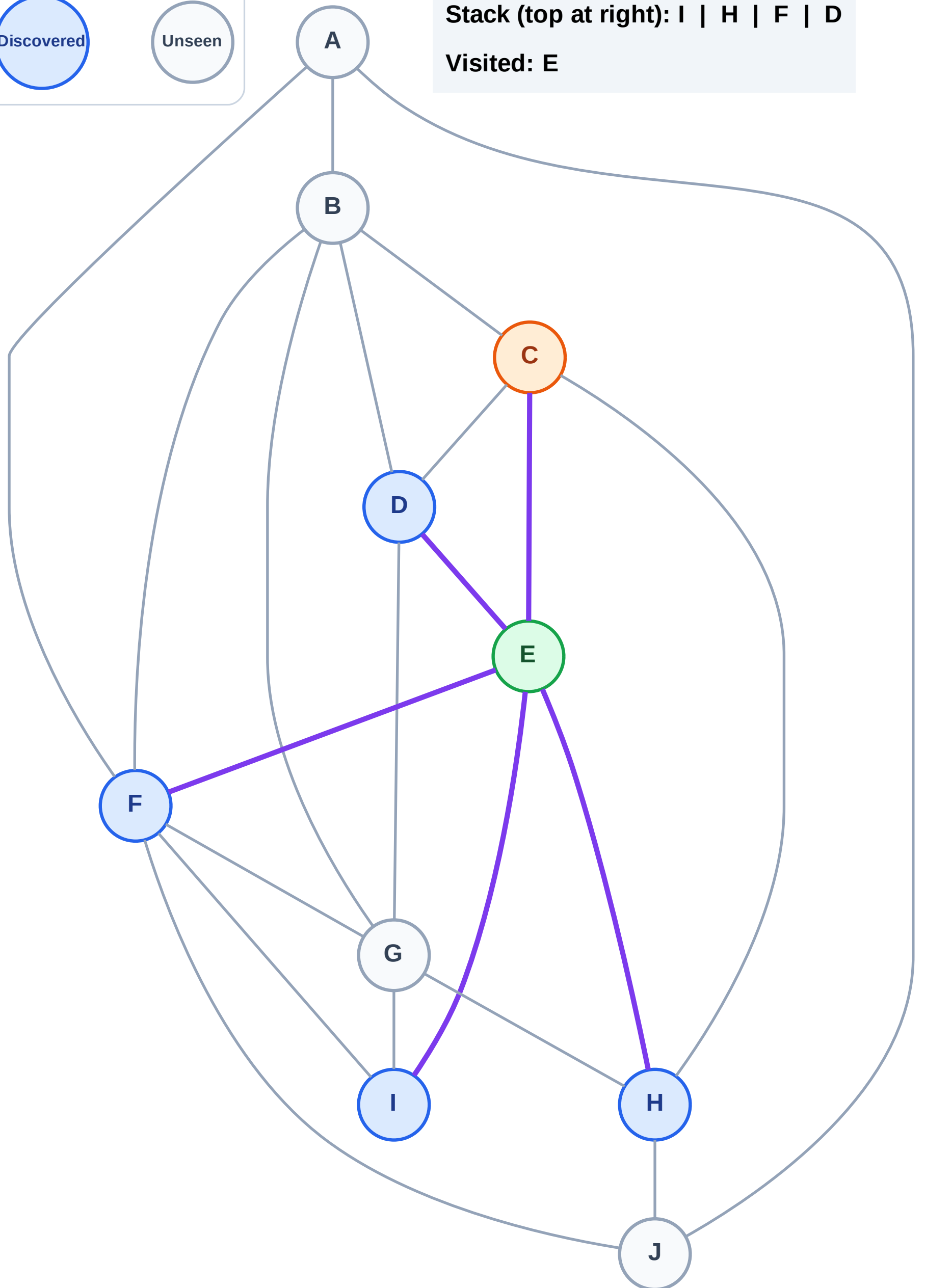
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D

Visited: E



DFS Traversal

Step 5: Discover B from C

Legend

Complete

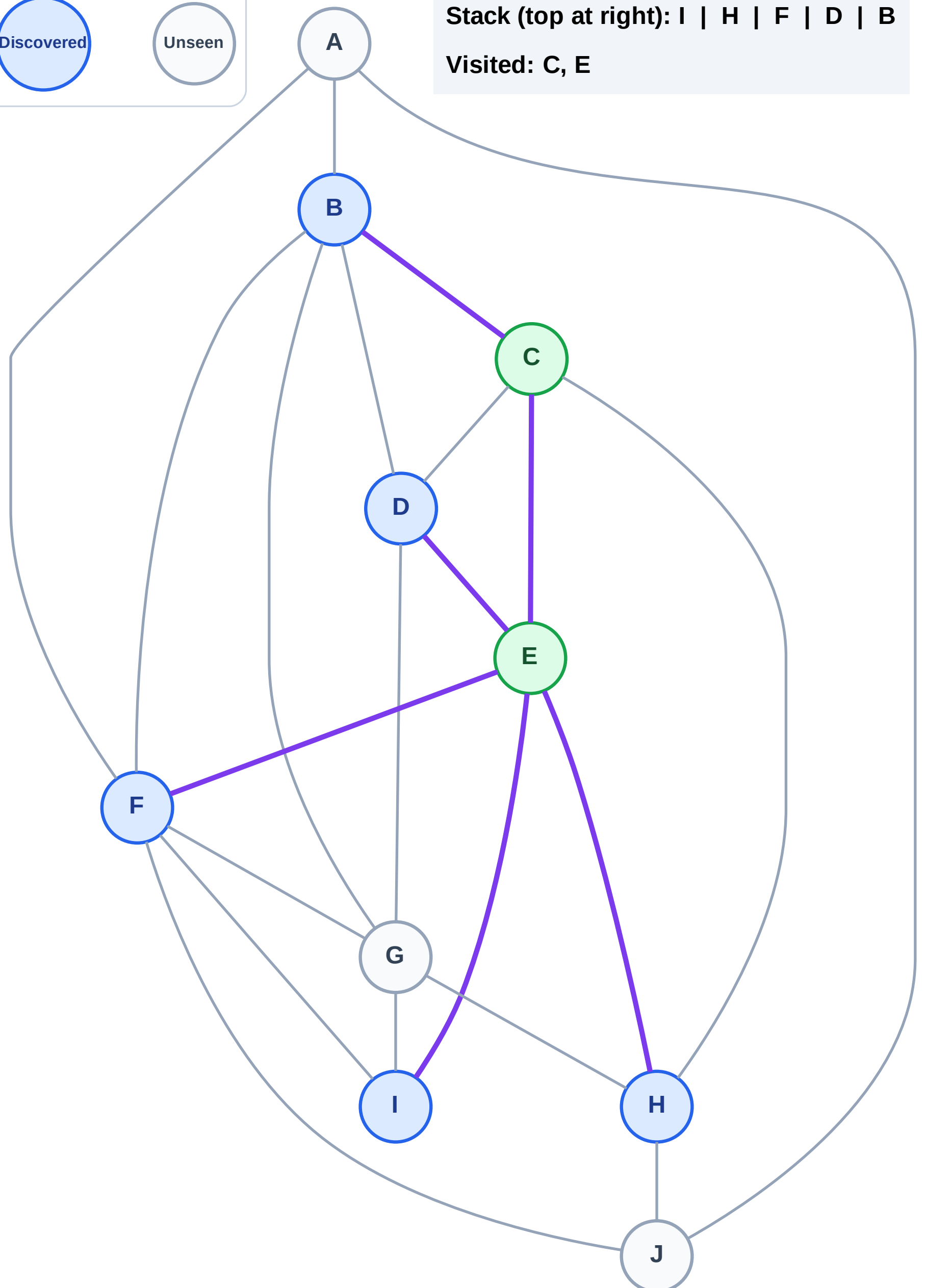
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | B

Visited: C, E



DFS Traversal

Step 6: Process node B

Legend

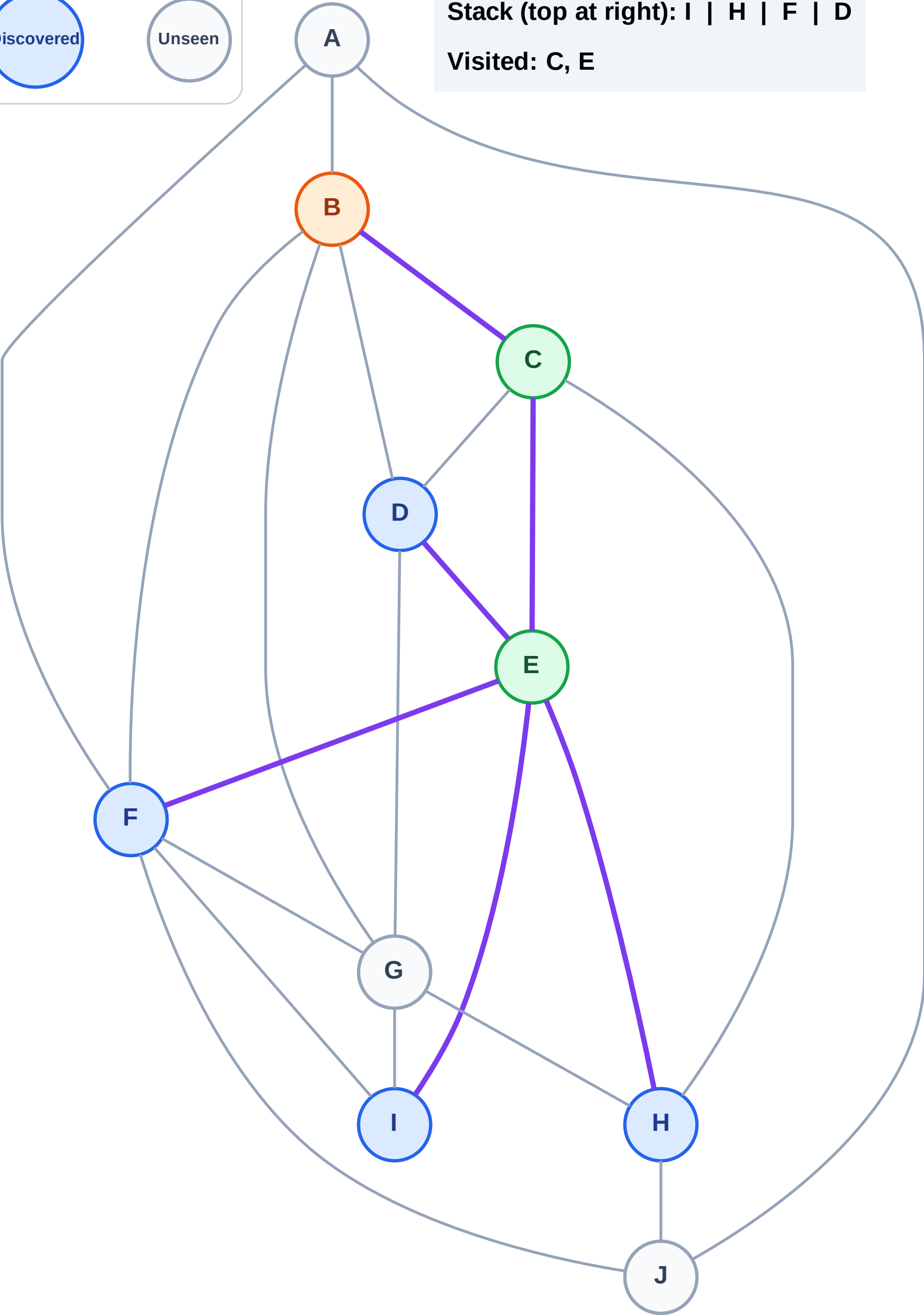
Complete

Processing

Discovered

Unseen

Stack (top at right): I | H | F | D
Visited: C, E



DFS Traversal

Step 7: Discover G, A from B

Legend

Complete

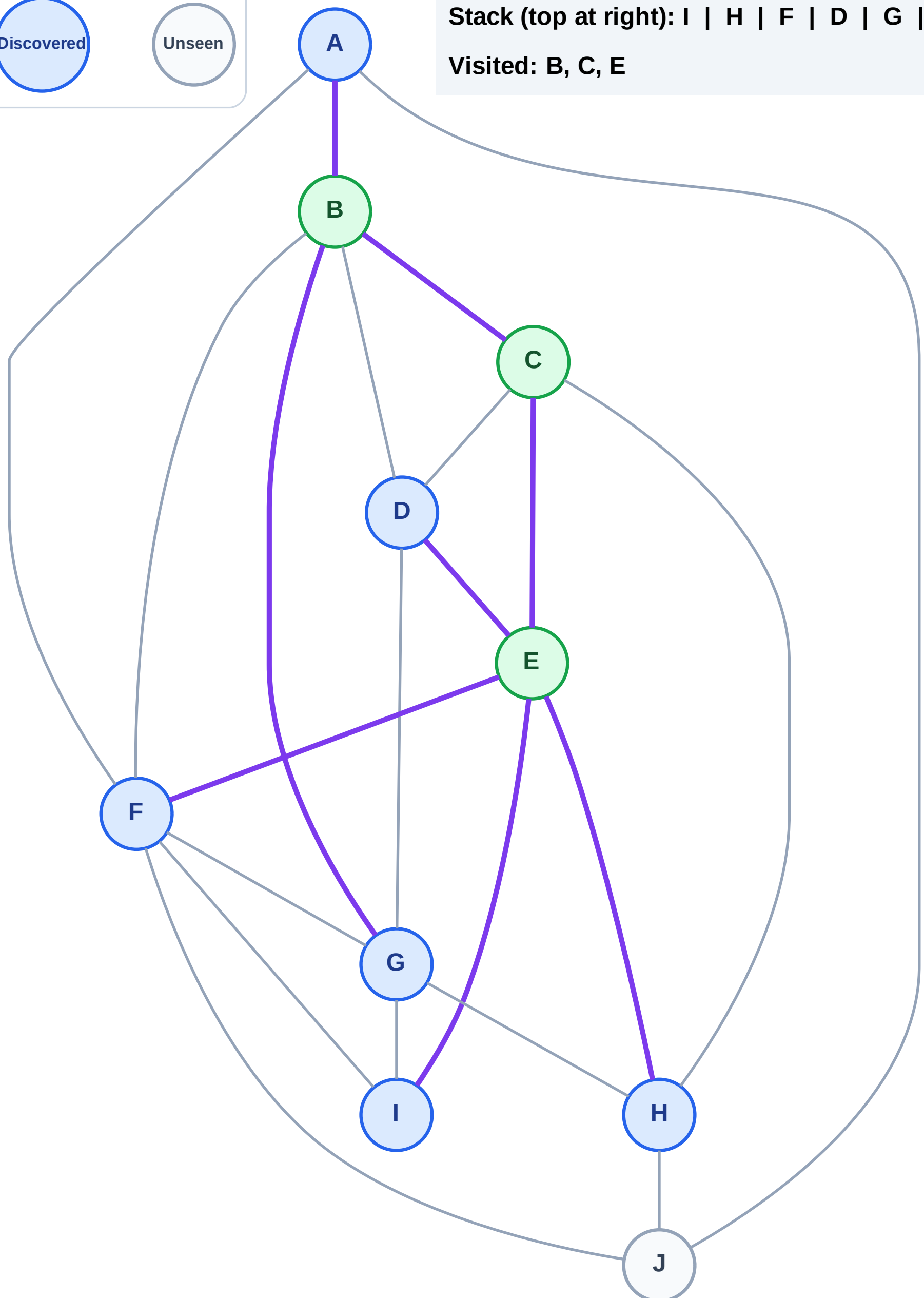
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | G | A

Visited: B, C, E



DFS Traversal

Step 8: Process node A

Legend

Complete

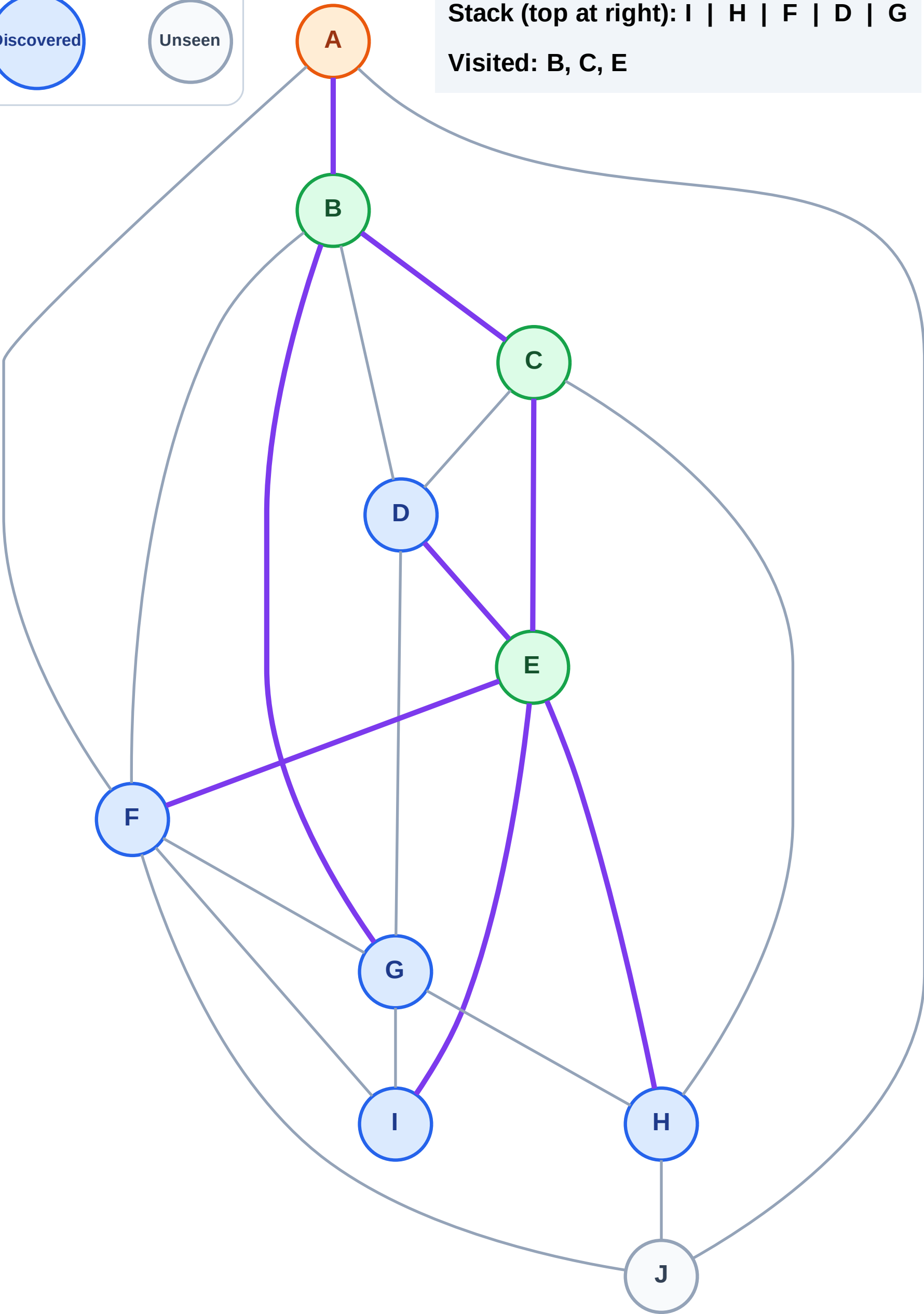
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | G

Visited: B, C, E



DFS Traversal

Step 9: Discover J from A

Legend

Complete

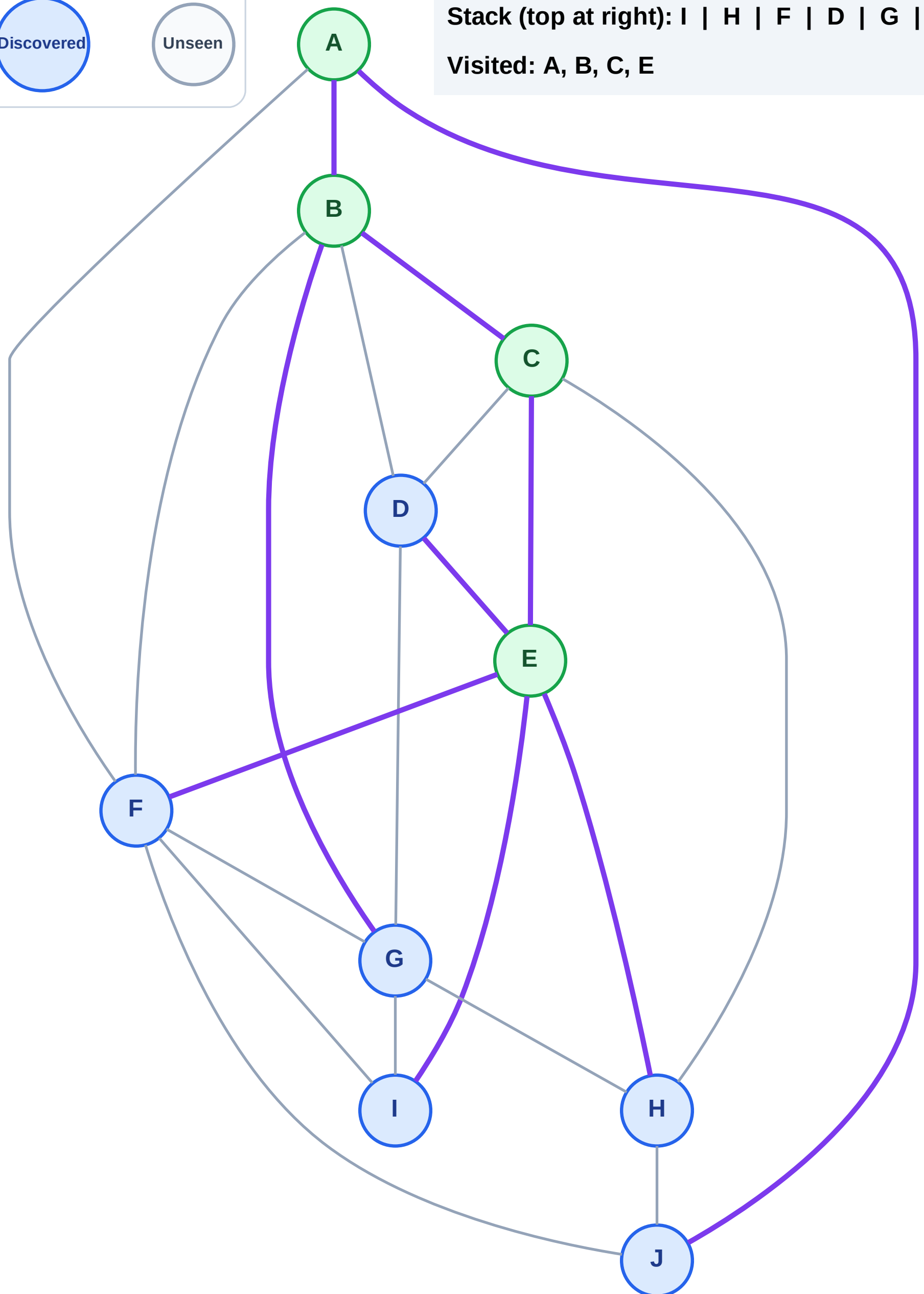
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | G | J

Visited: A, B, C, E



DFS Traversal

Step 10: Process node J

Legend

Complete

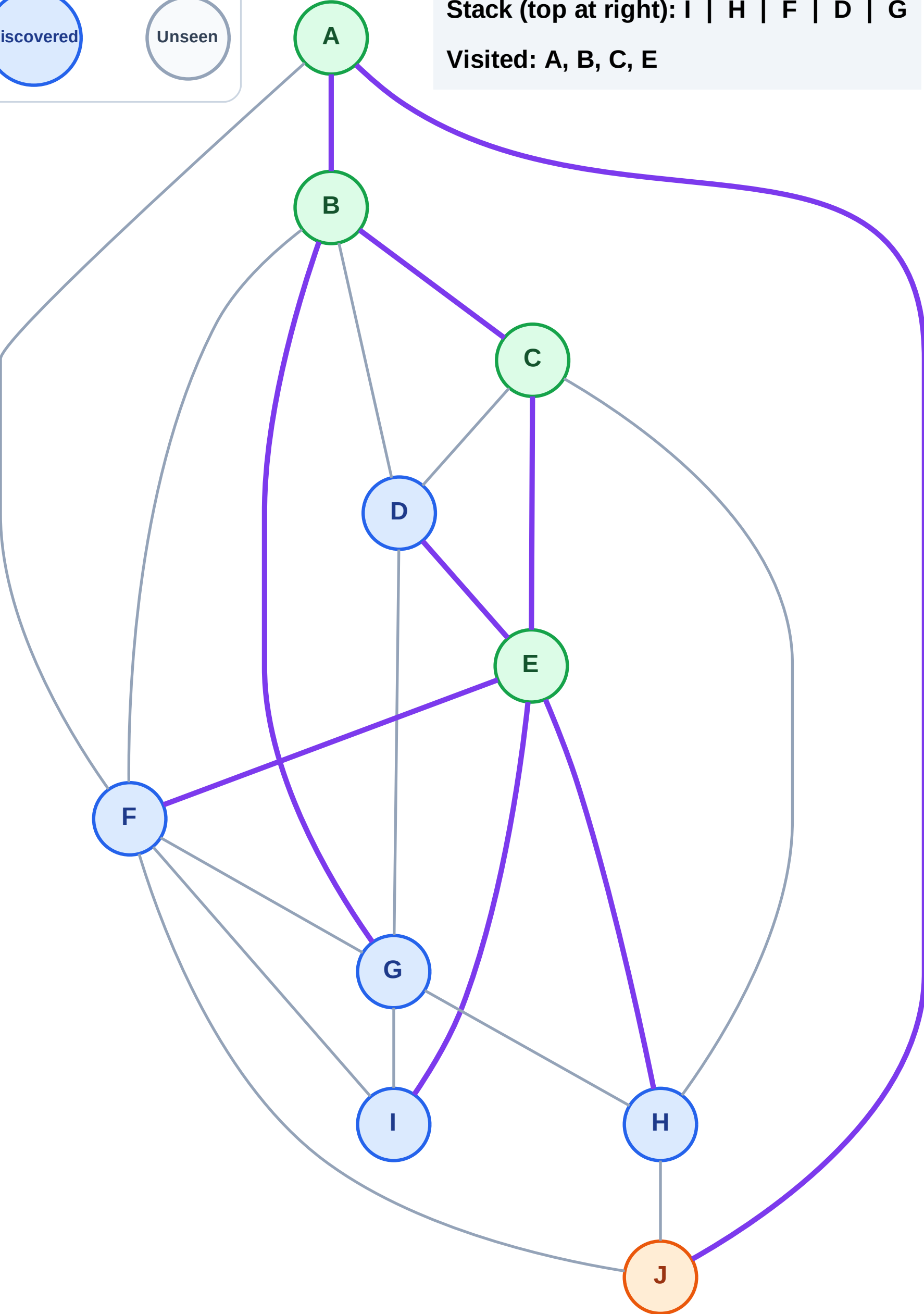
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | G

Visited: A, B, C, E



DFS Traversal

Step 11: No new neighbors from J

Legend

Complete

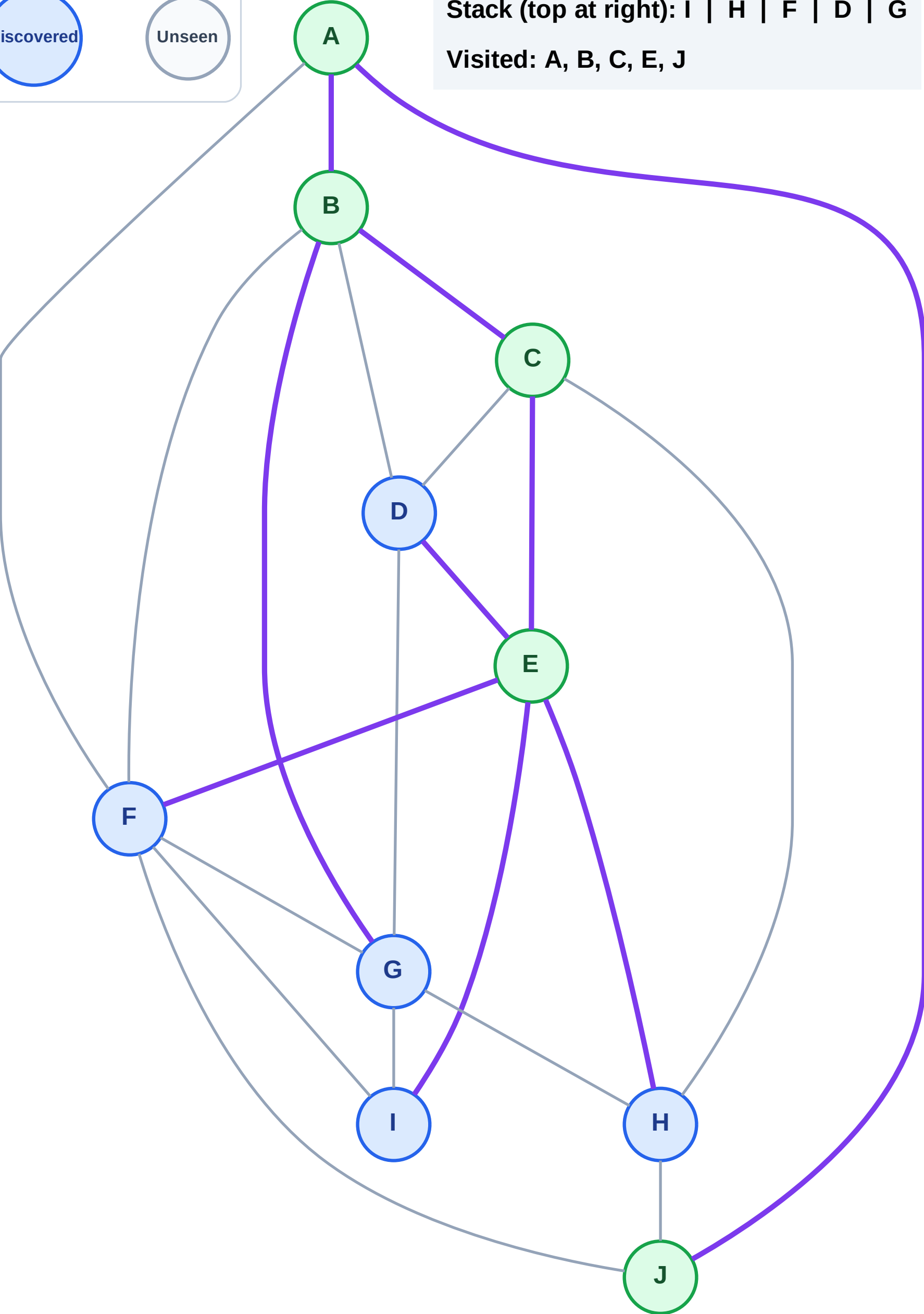
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | G

Visited: A, B, C, E, J



DFS Traversal

Step 12: Process node G

Legend

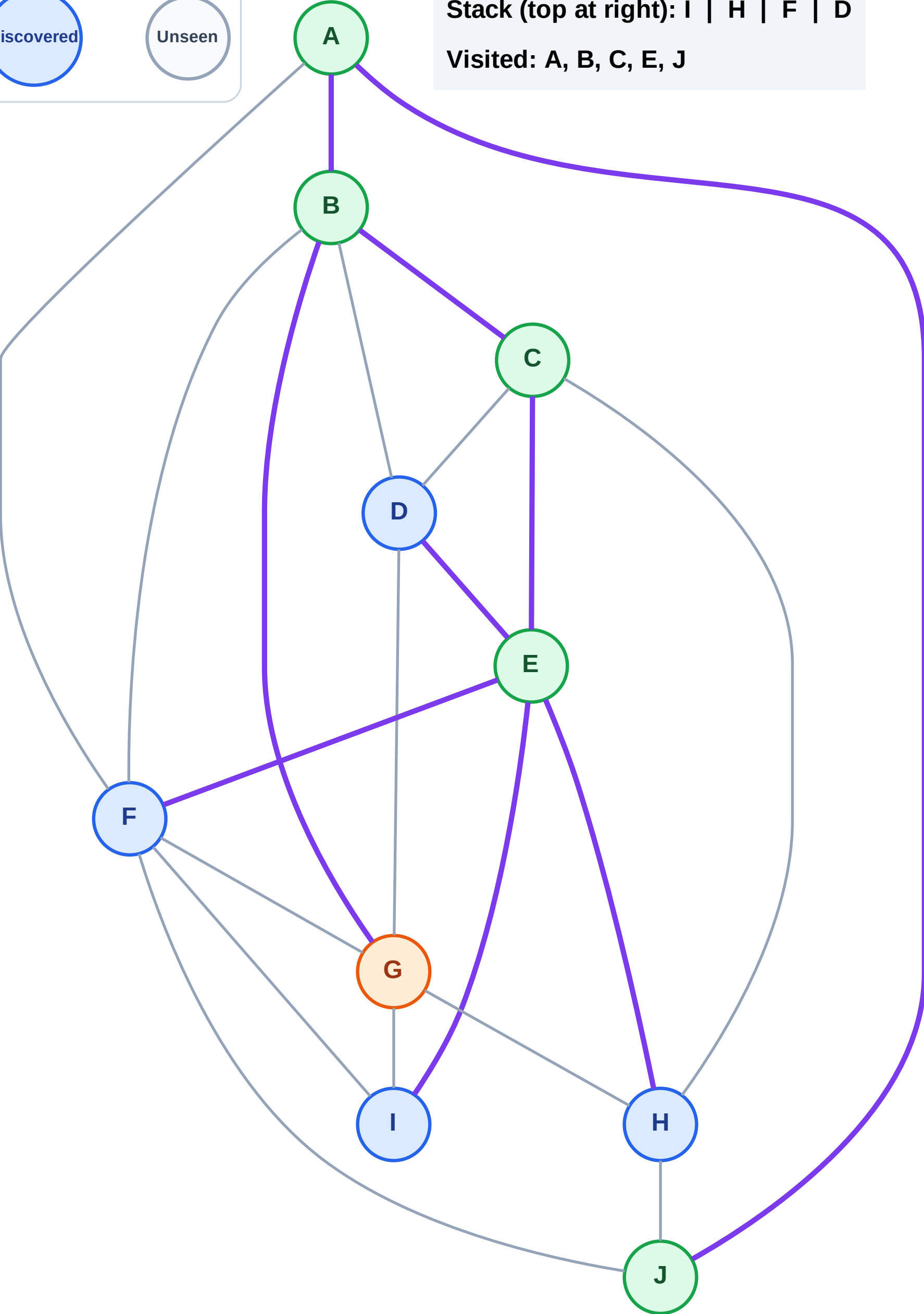
Complete

Processing

Discovered

Unseen

Stack (top at right): I | H | F | D
Visited: A, B, C, E, J



DFS Traversal

Step 13: No new neighbors from G

Legend

Complete

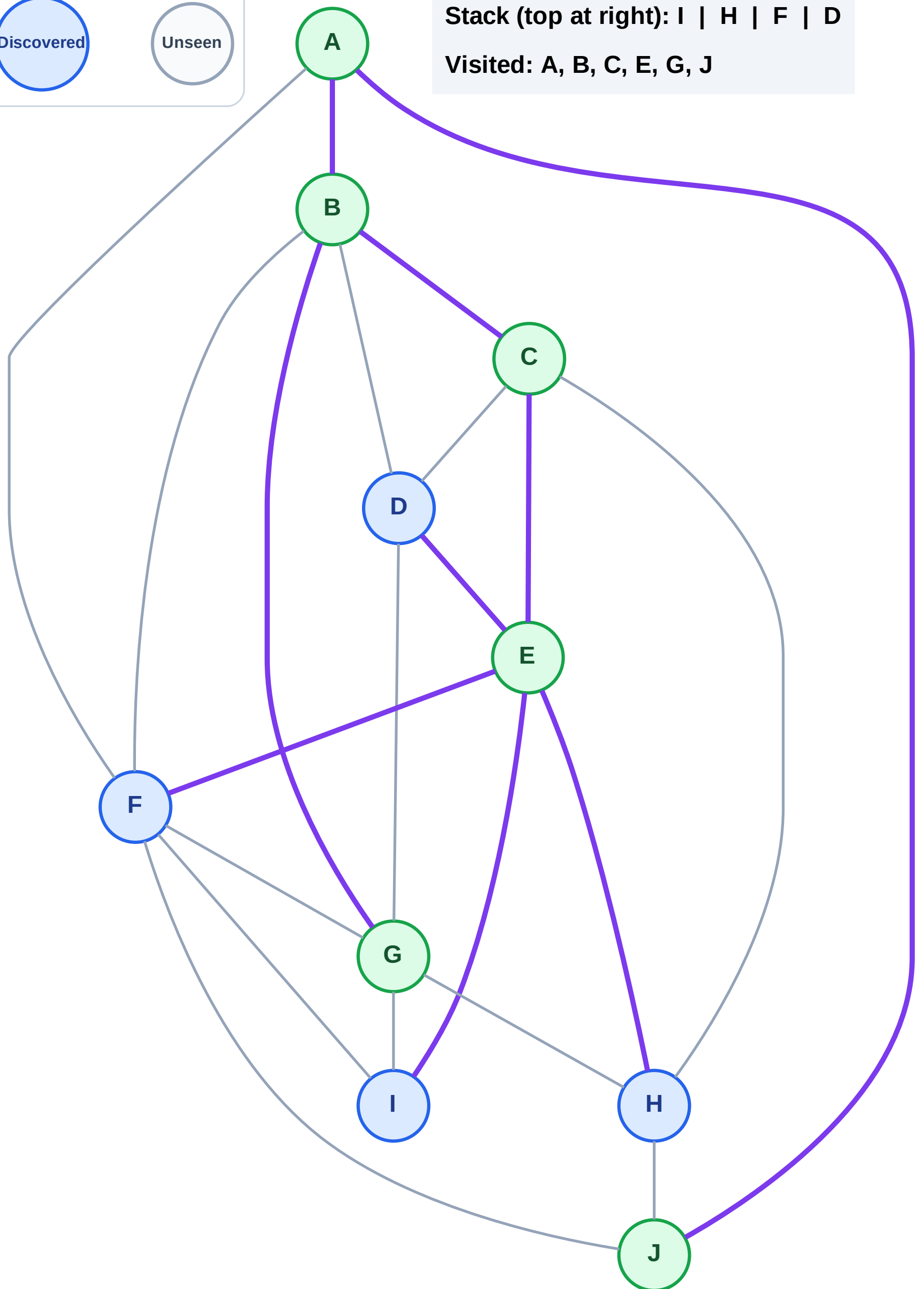
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D

Visited: A, B, C, E, G, J



DFS Traversal

Step 14: Process node D

Legend

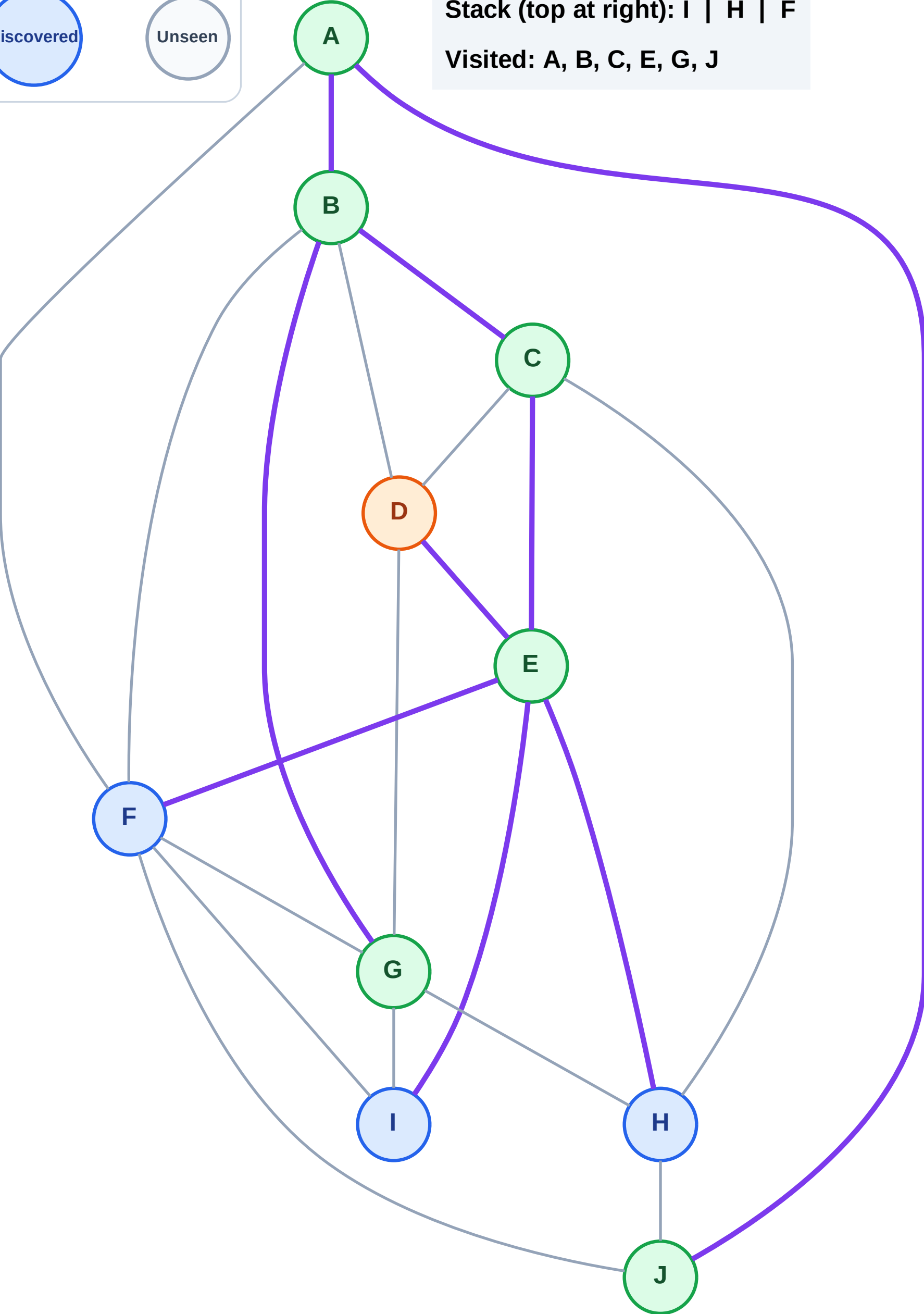
Complete

Processing

Discovered

Unseen

Stack (top at right): I | H | F
Visited: A, B, C, E, G, J



DFS Traversal

Step 15: No new neighbors from D

Legend

Complete

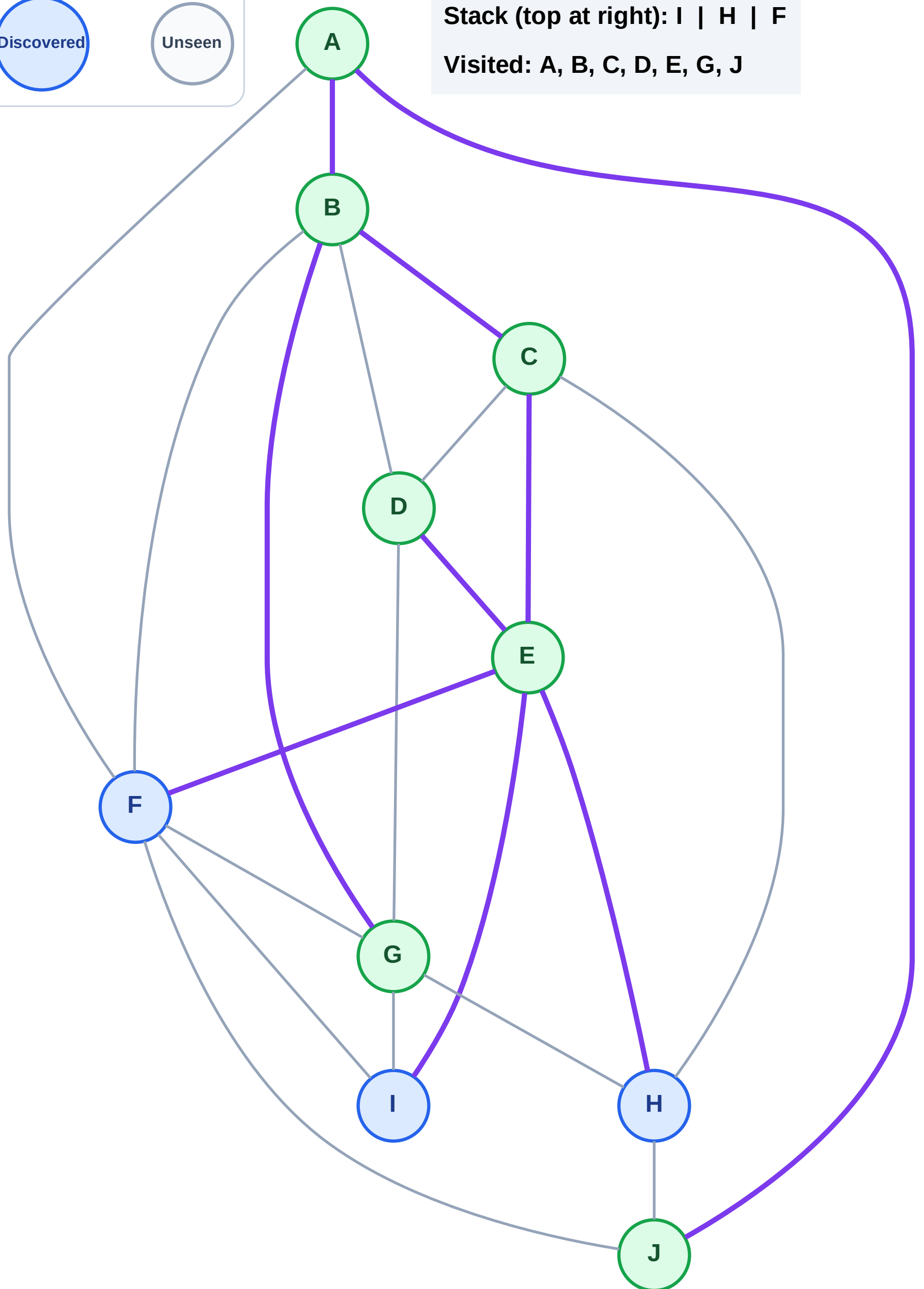
Processing

Discovered

Unseen

Stack (top at right): I | H | F

Visited: A, B, C, D, E, G, J



DFS Traversal

Step 16: Process node F

Stack (top at right): I | H
Visited: A, B, C, D, E, G, J

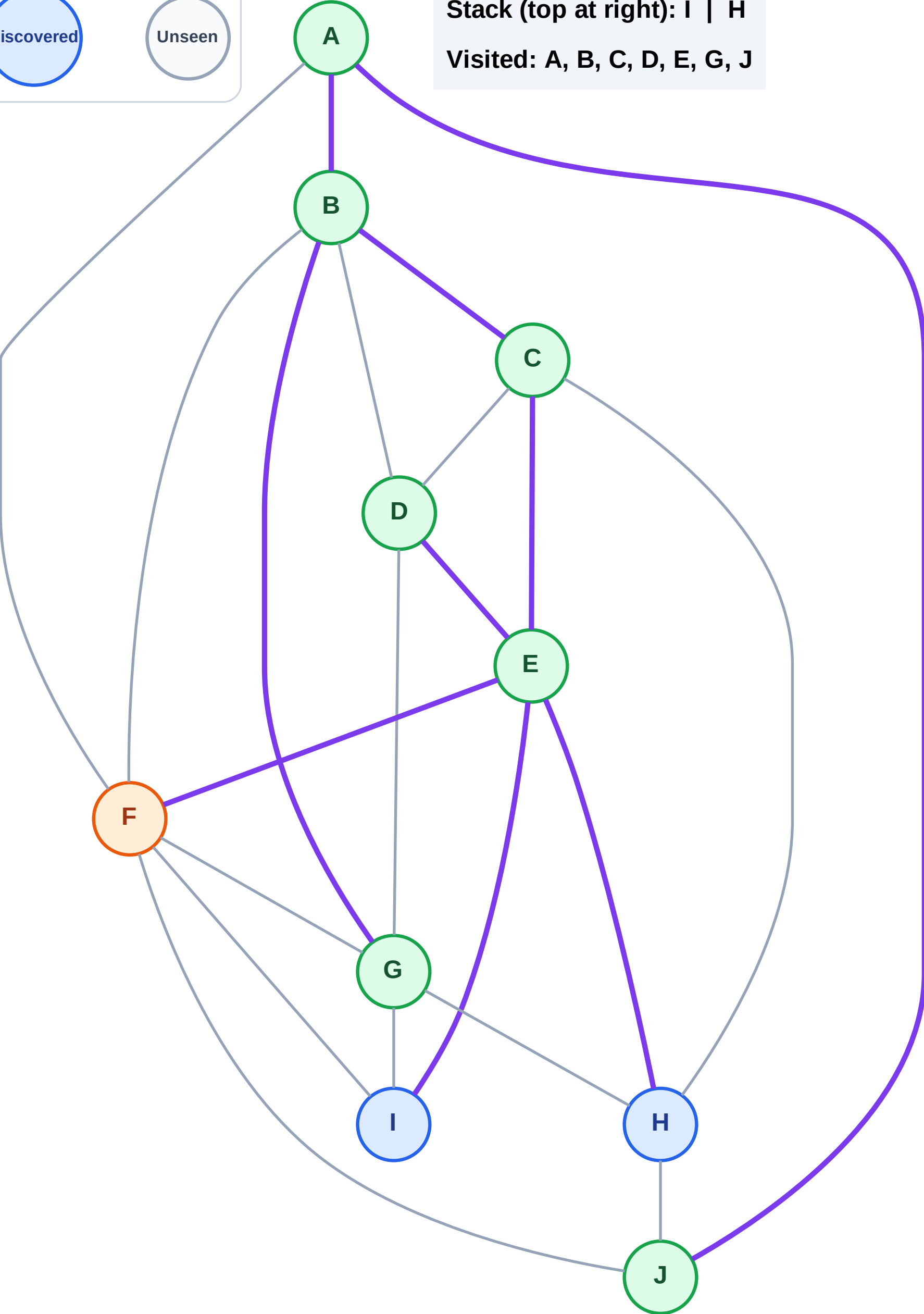
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 17: No new neighbors from F

Legend

Complete

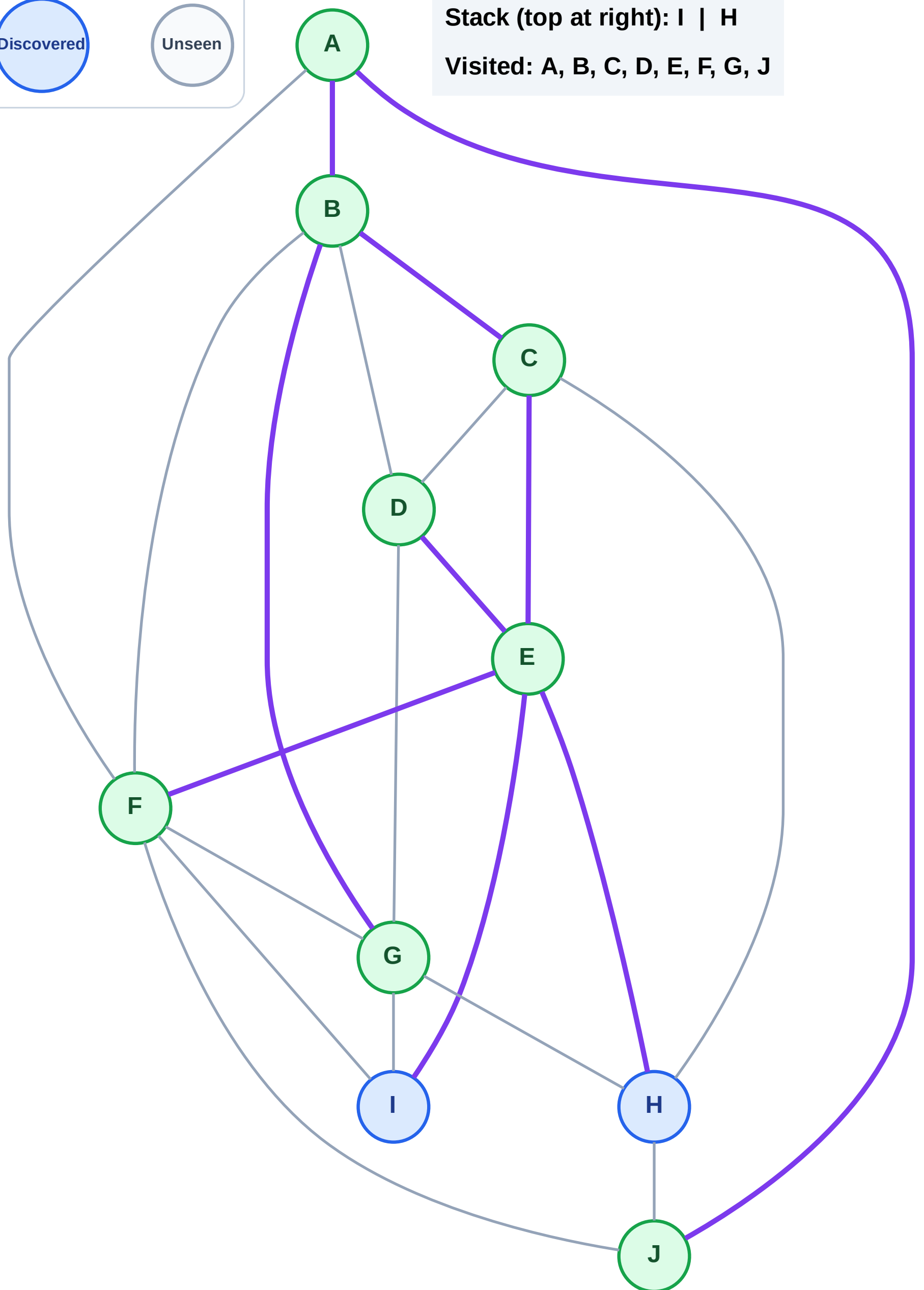
Processing

Discovered

Unseen

Stack (top at right): I | H

Visited: A, B, C, D, E, F, G, J



DFS Traversal

Step 18: Process node H

Legend

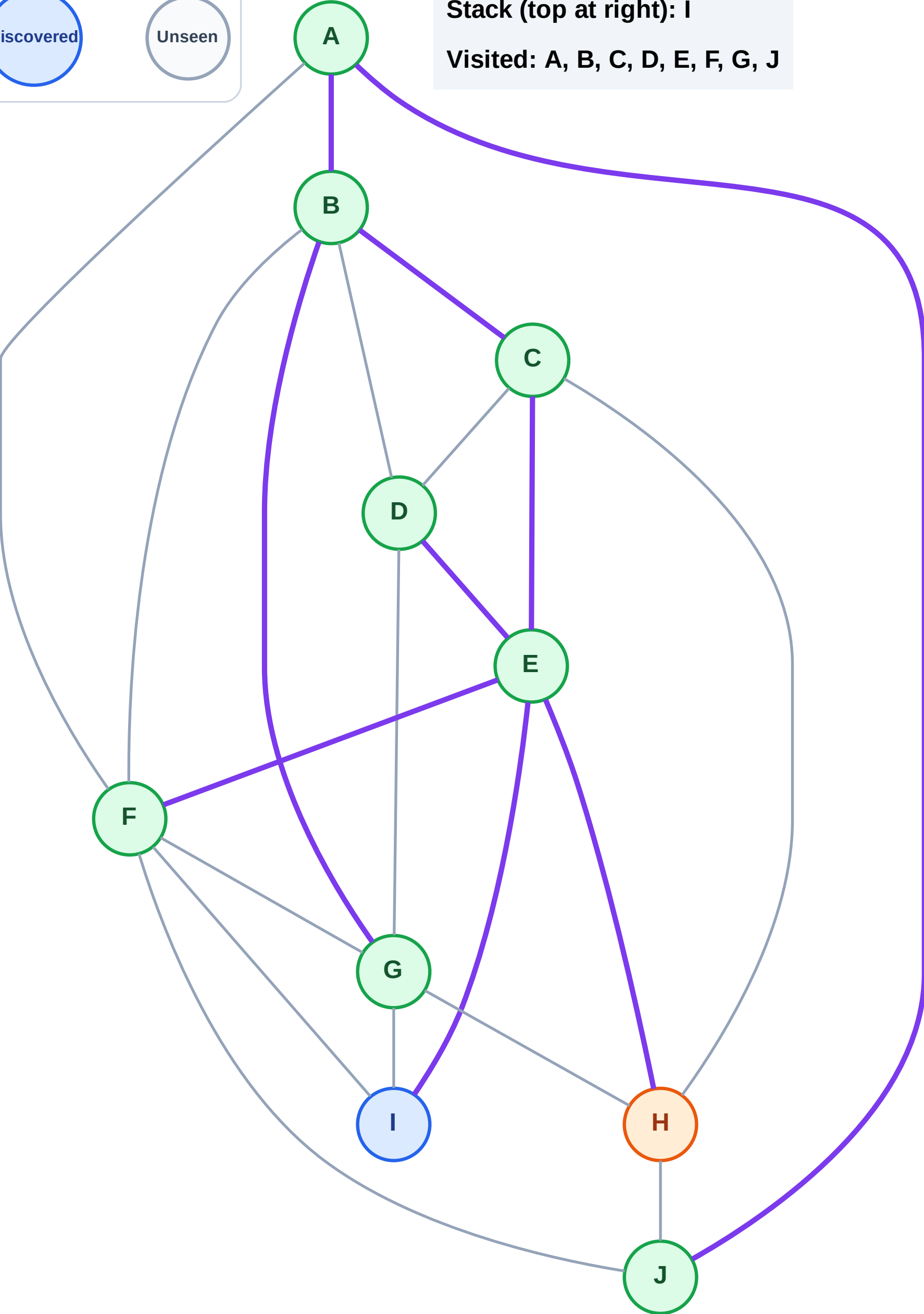
Complete

Processing

Discovered

Unseen

Stack (top at right): I
Visited: A, B, C, D, E, F, G, J



DFS Traversal

Step 19: No new neighbors from H

Legend

Complete

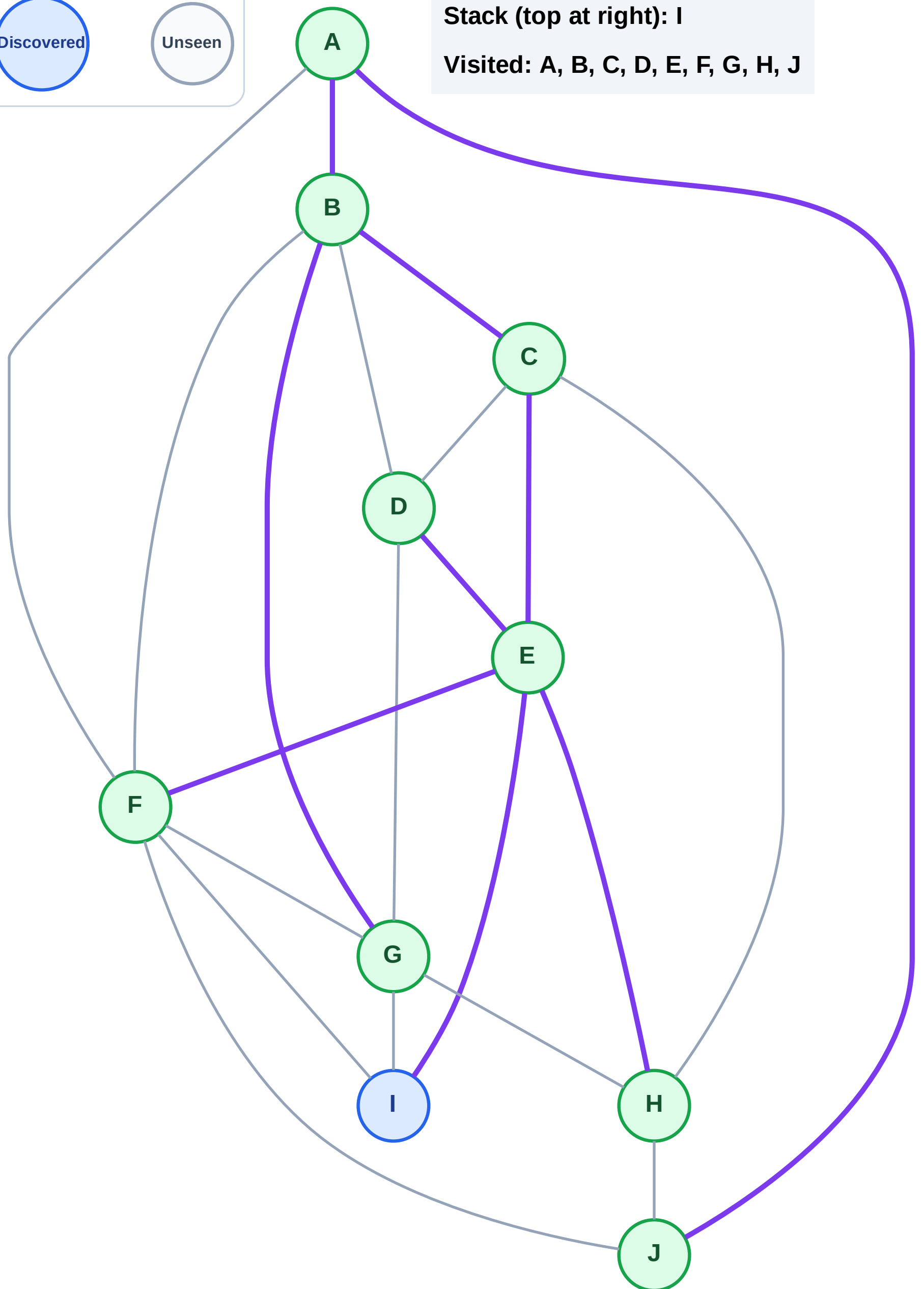
Processing

Discovered

Unseen

Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

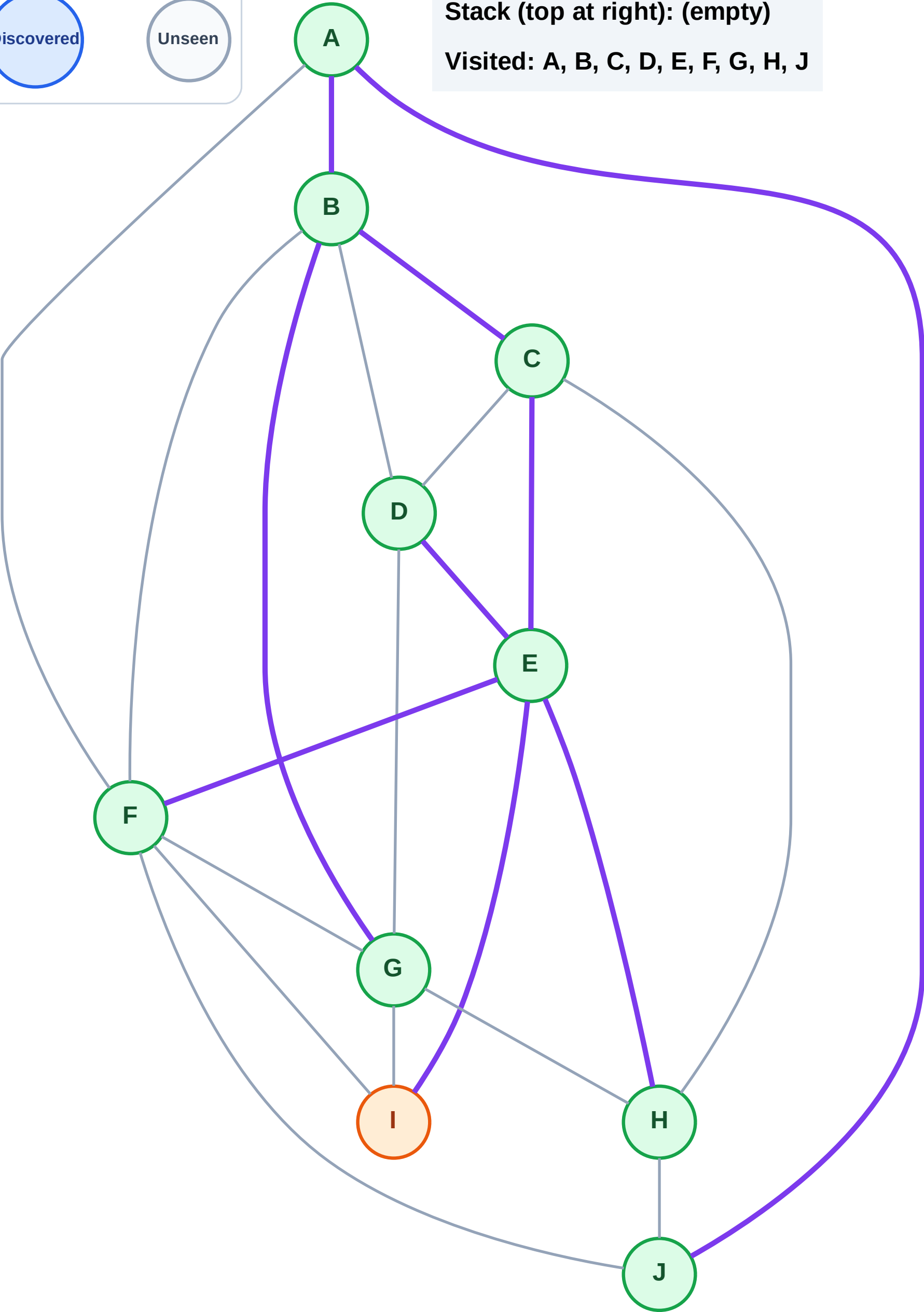
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

